生物与电子

王小勤

清华大学生物医学工程系 教授
清华大学脑与智能实验室 主任

我的教育经历

- 本科：电子工程（无线电系）
- 硕士：电子工程及计算机科学
- 博士：生物医学工程
- 研究方向：脑科学、神经工程

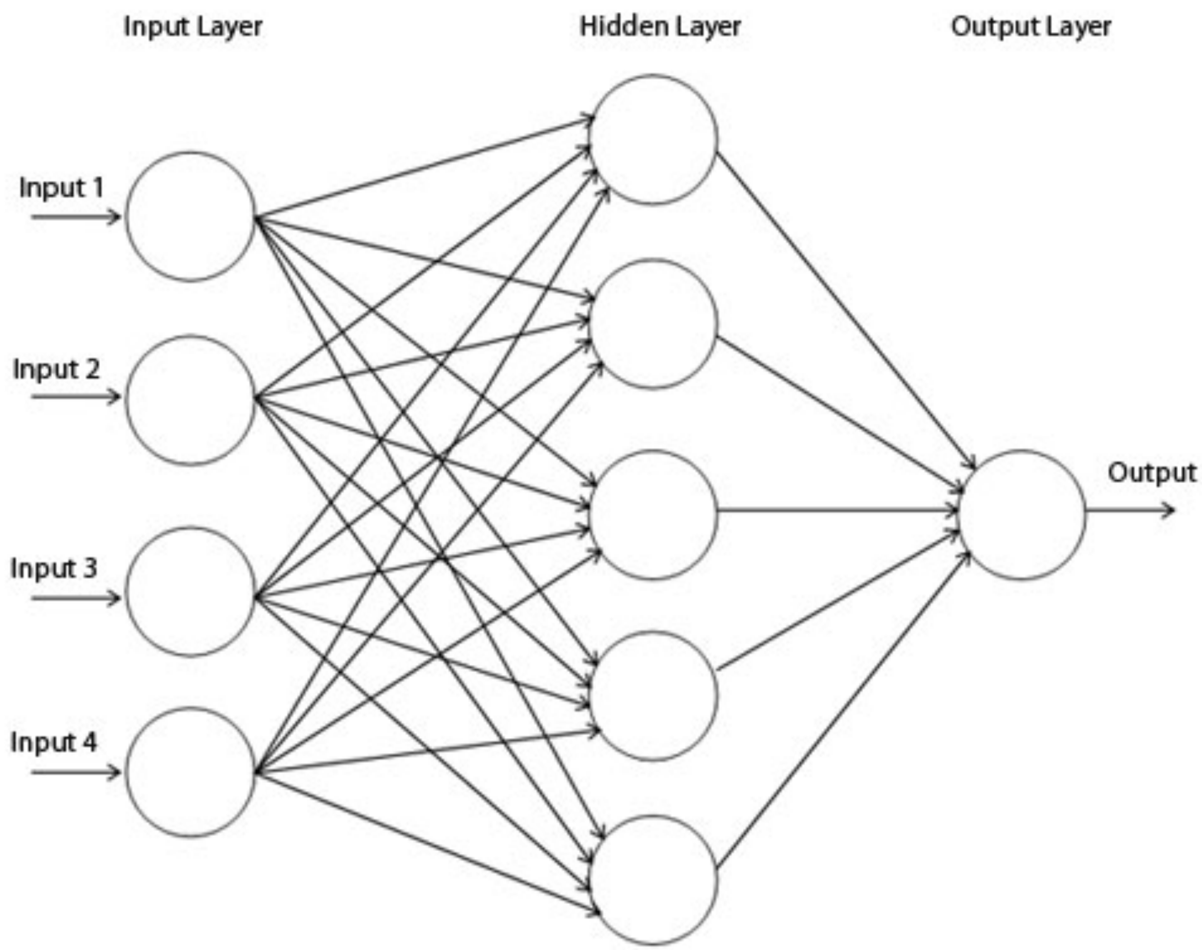
2018年图灵奖获得者

最新一届图灵奖被授予了 "深度学习三巨头" ：约书亚·本希奥（Yoshua Bengio）、杰弗里·欣顿（Geoffrey Hinton）和杨立昆（Yann LeCun）|图片来源：Botler AI

什么是“人工智能”？


VS
nature
At last — a computer program that
can beat a champion Go player
ALL SYSTEMS GO

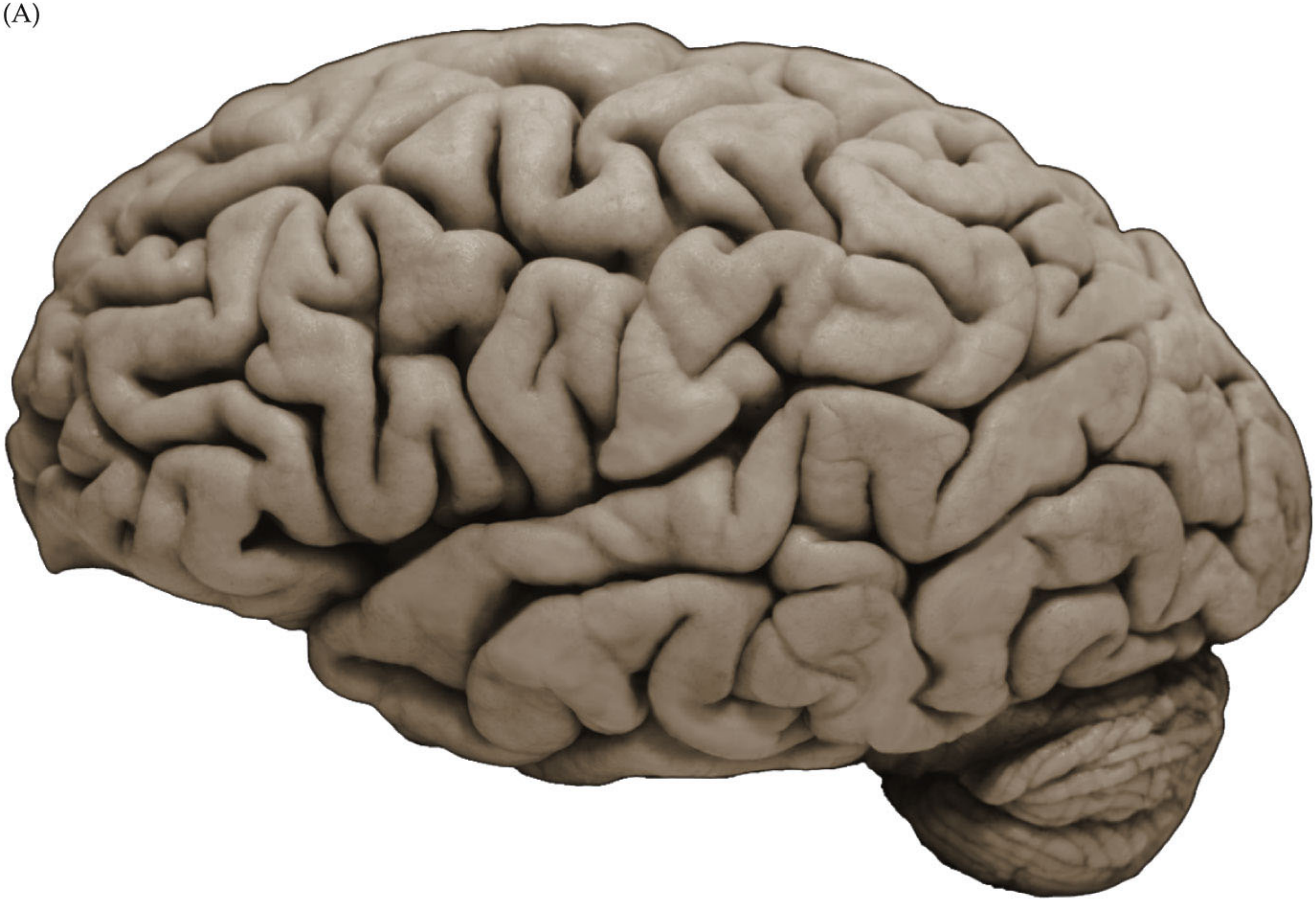
神经网络 （大数据，深度学习）



从“阿尔法狗”到通用人工智能

The Human Brain

(A)



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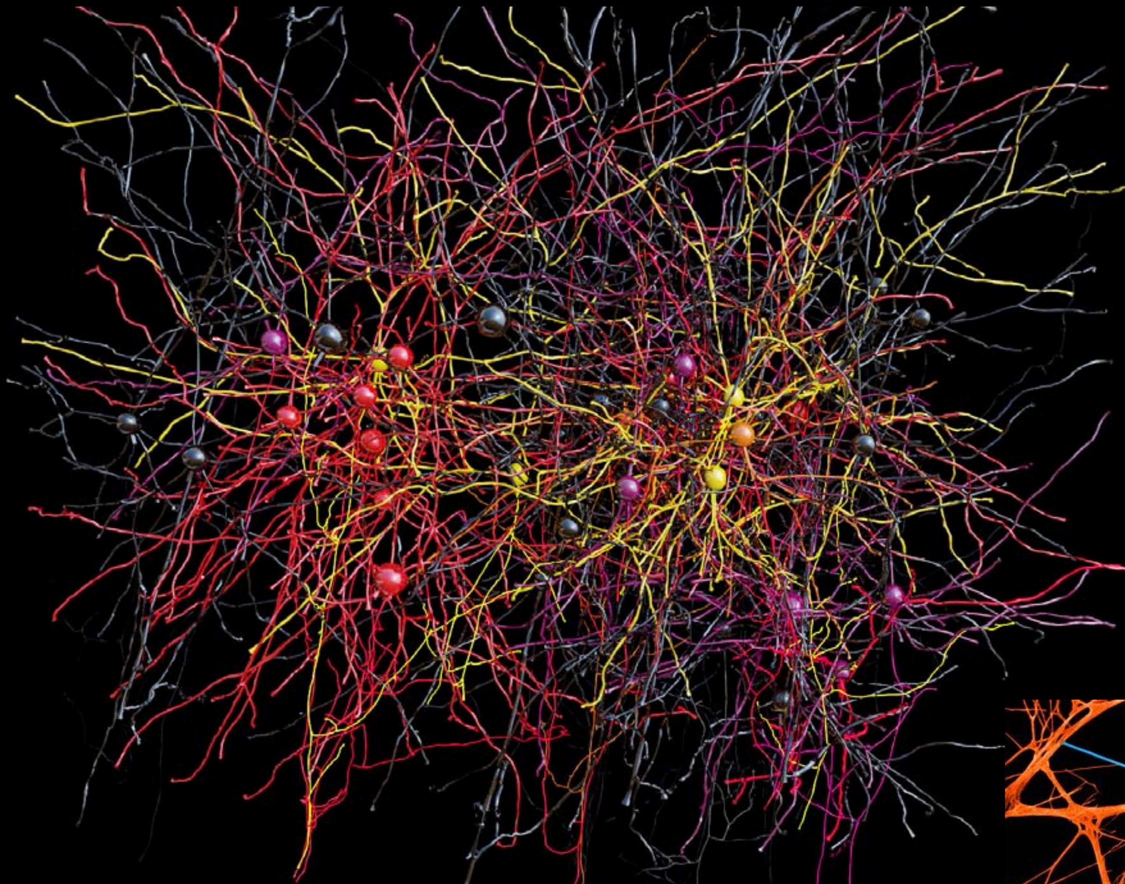
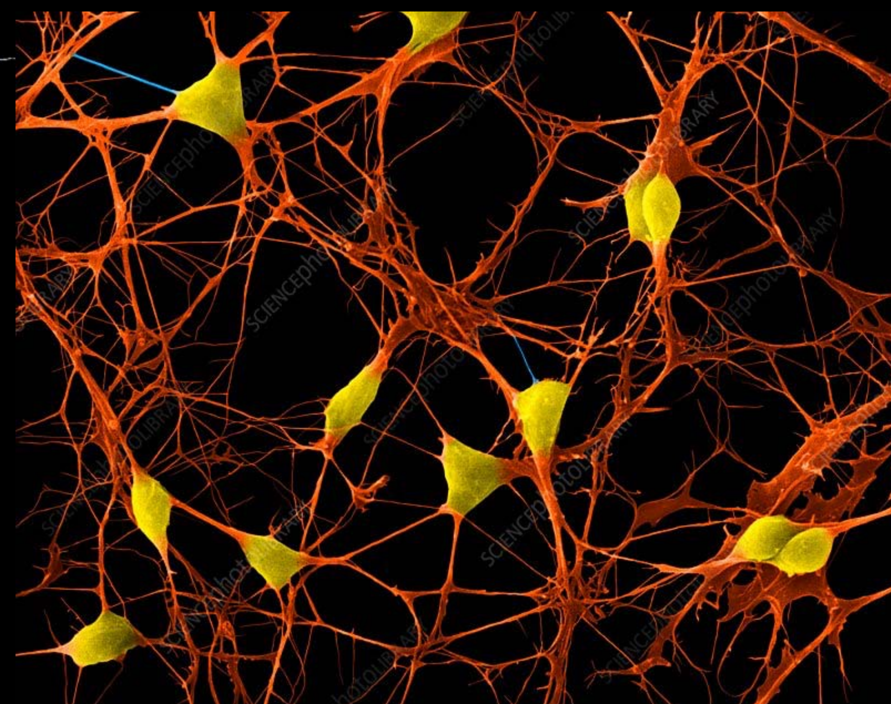
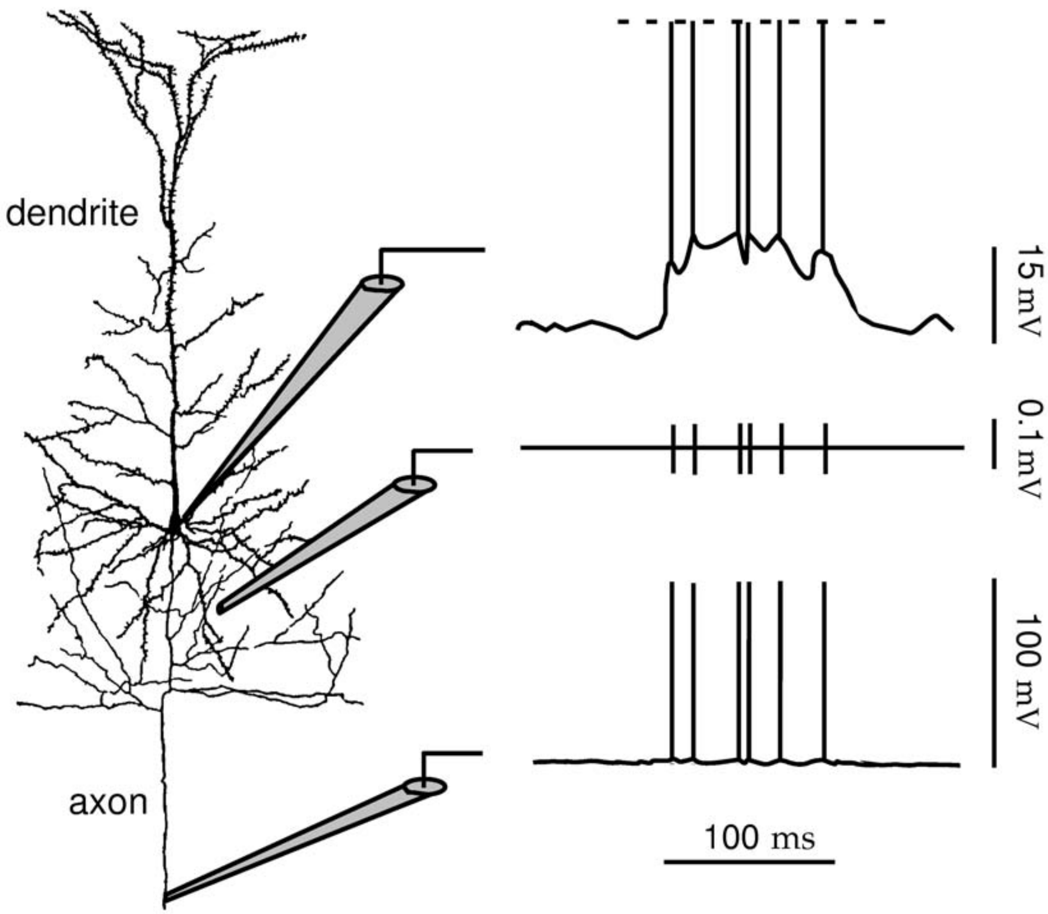
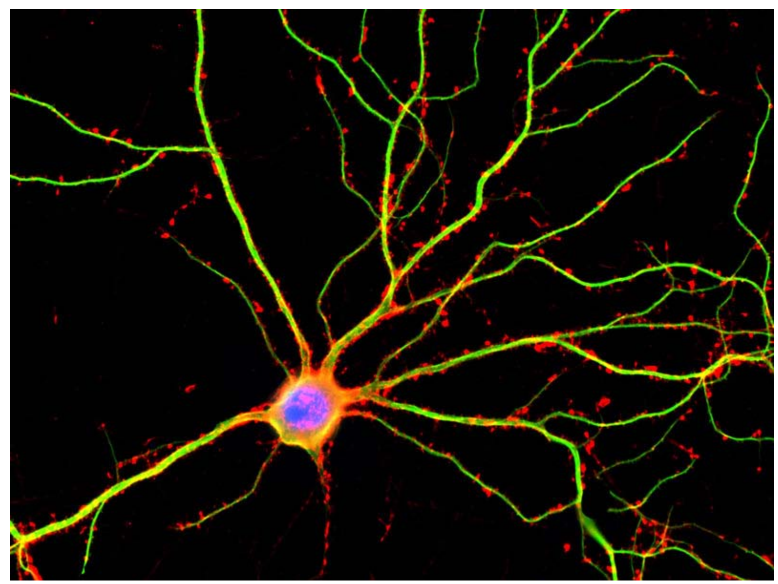


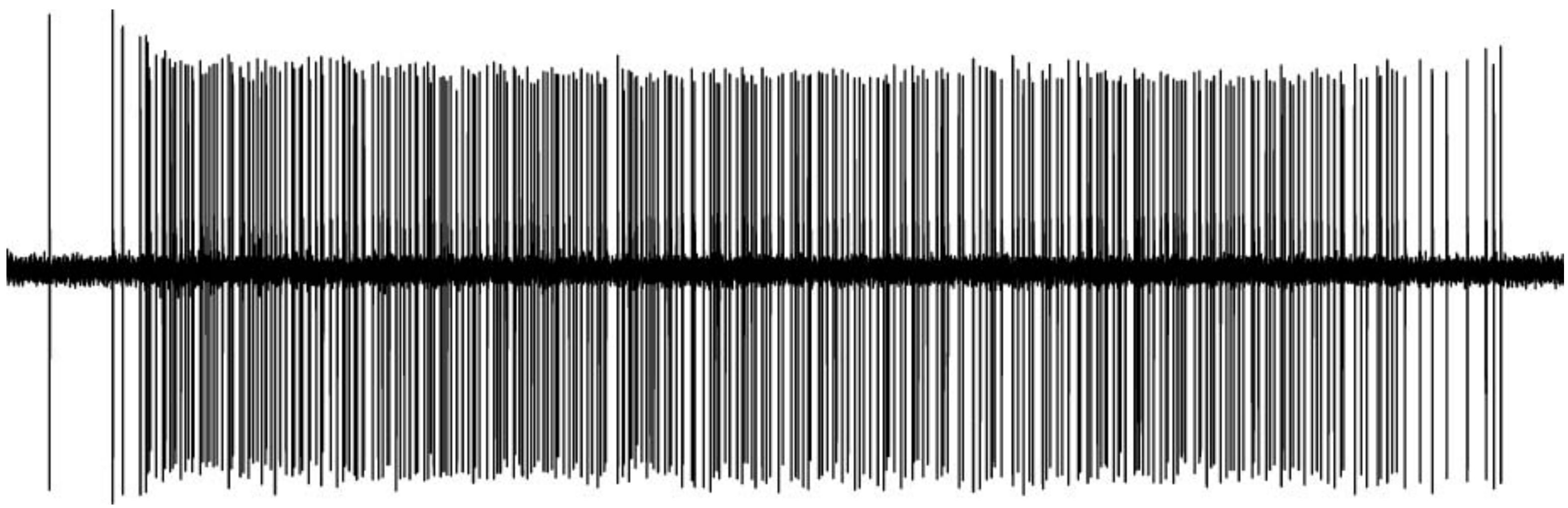


大脑神经元

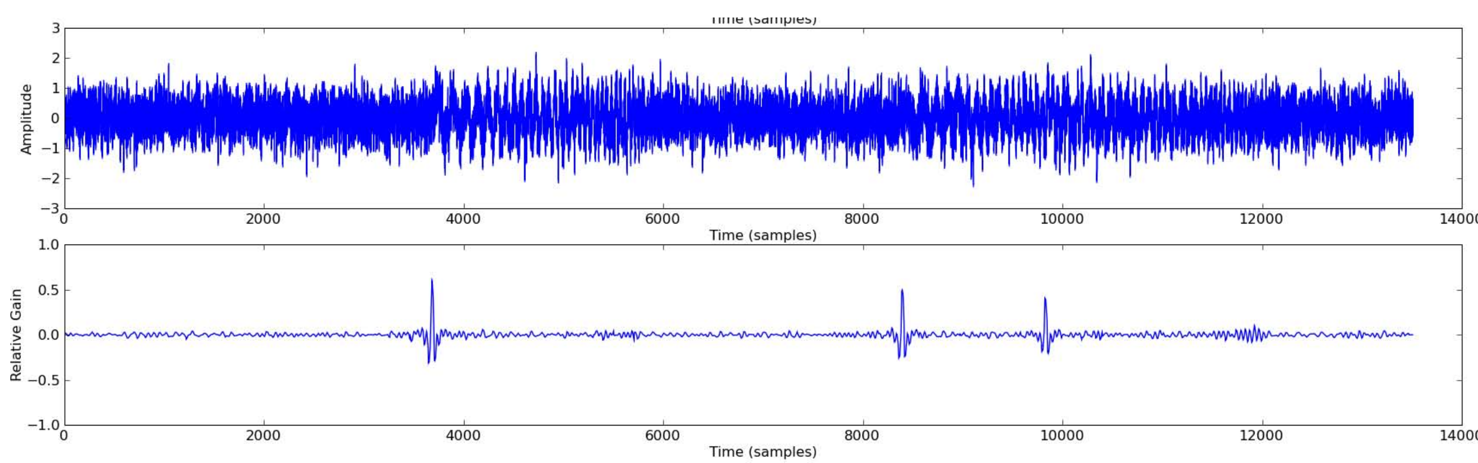




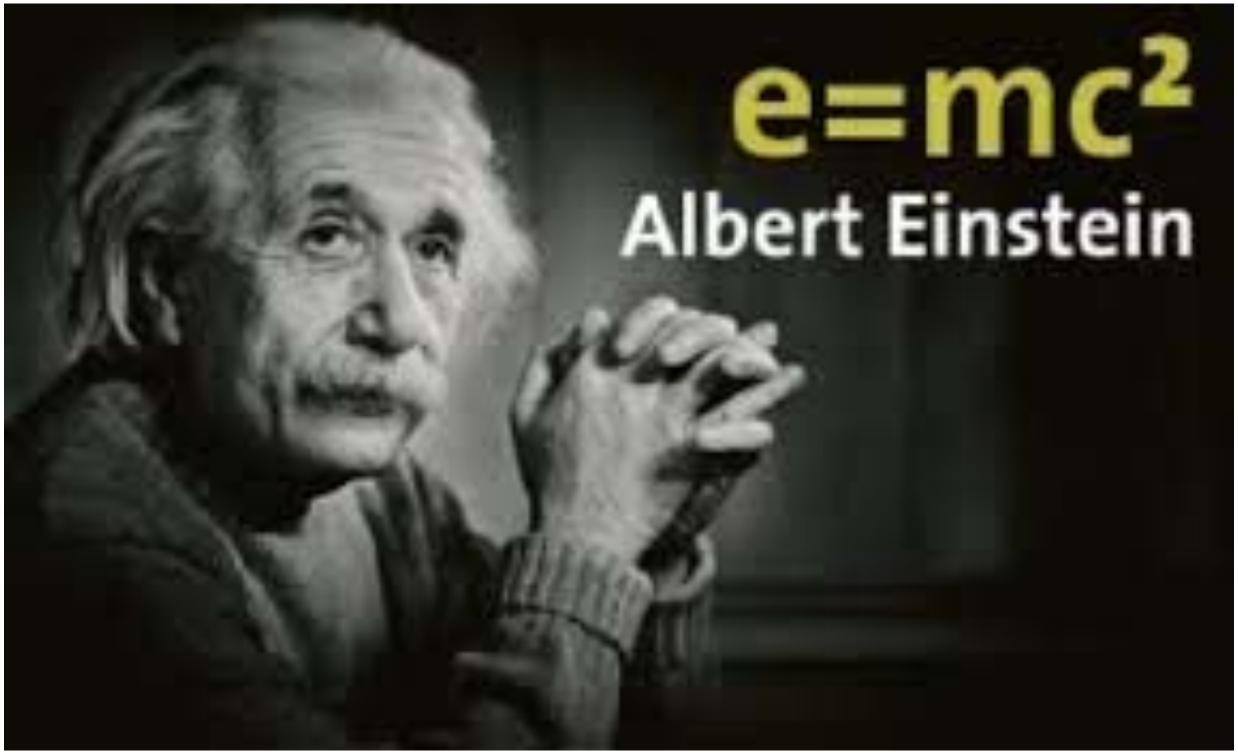
大脑神经元传输信息的方式：神经电脉冲



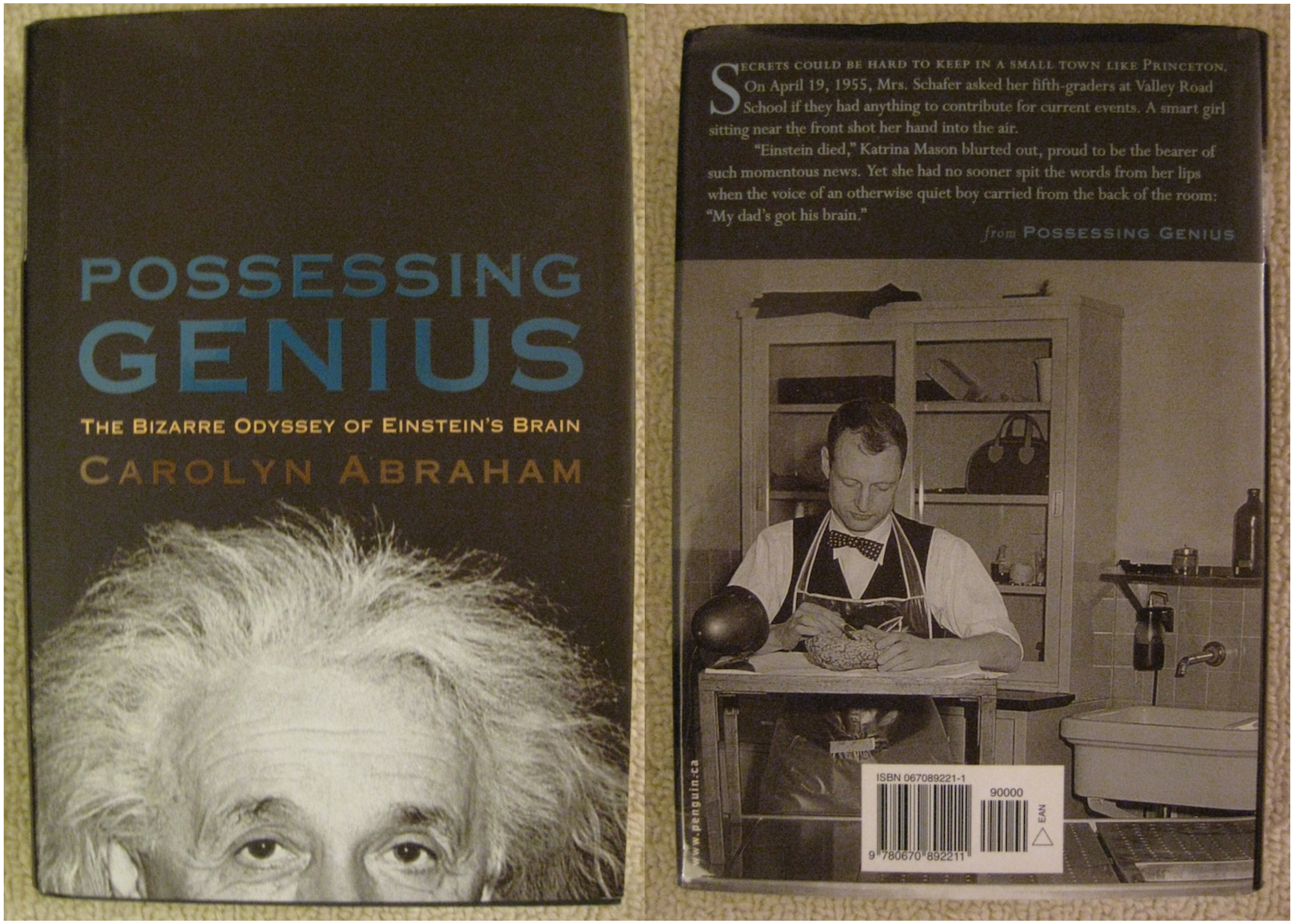
雷达信号



谁的大脑你最想了解？

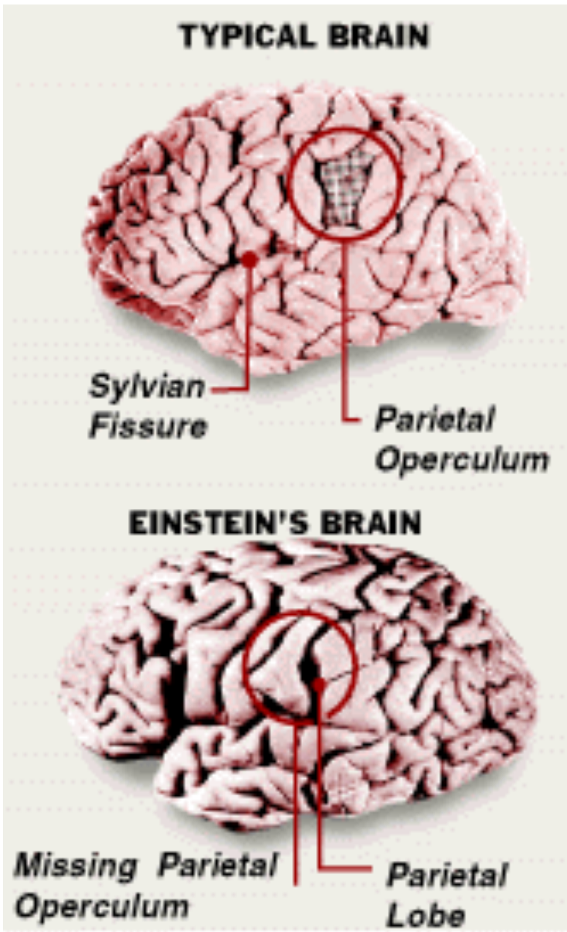


爱因斯坦大脑中的秘密是什么？



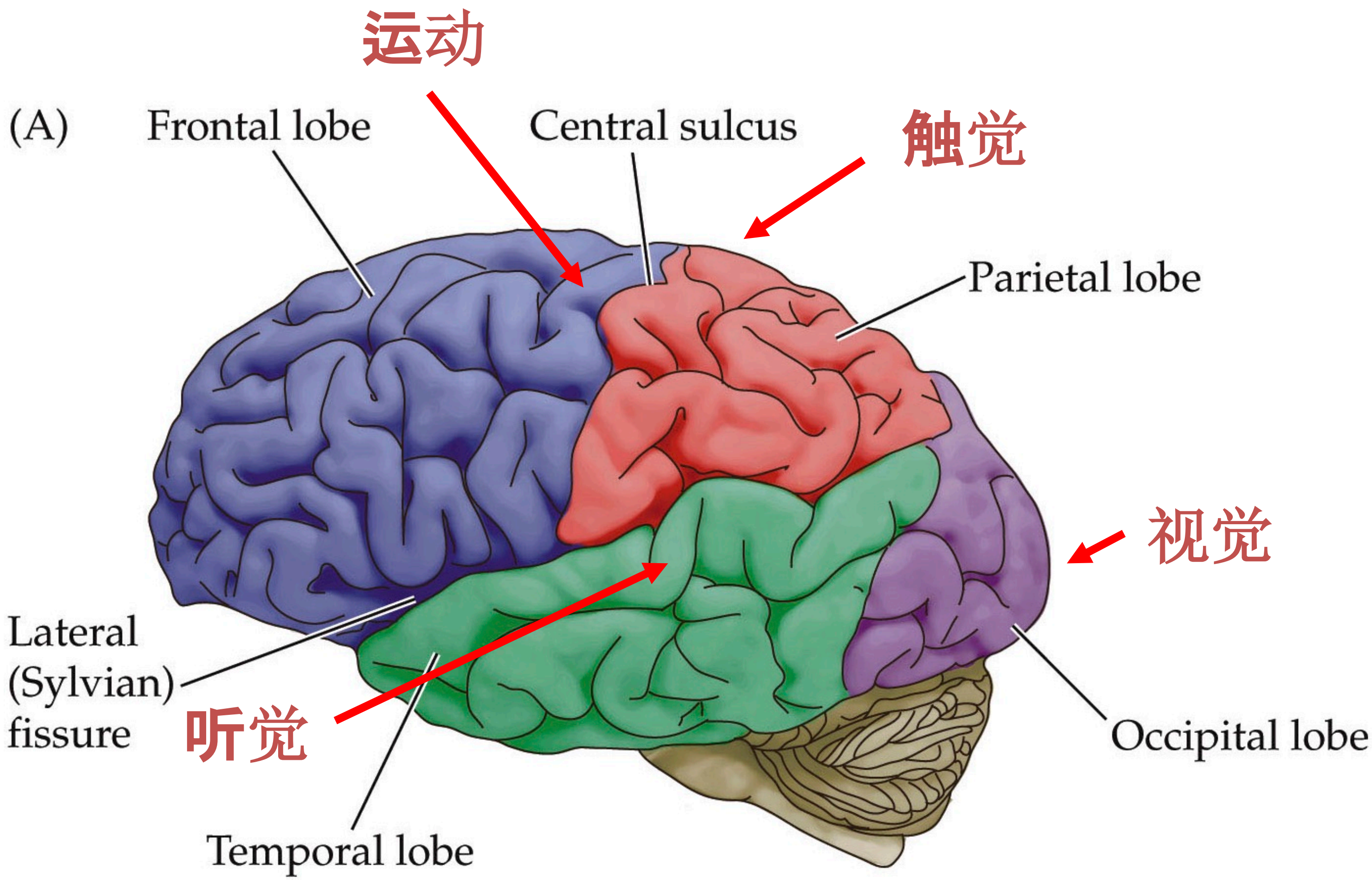
"For nearly half a century the pickled brain of Albert Einstein has roamed the world in Tupperware containers, courier packages, and, most famously, car trunks." (From *Possessing Genius: The Bizarre Odyssey of Einstein's Brain*, by Carolyn Abraham)

爱因斯坦大脑特殊之处在哪？



Einstein's brain weight was not different from that of controls, clearly indicating that a large (heavy) brain is not a necessary condition for exceptional intellect.

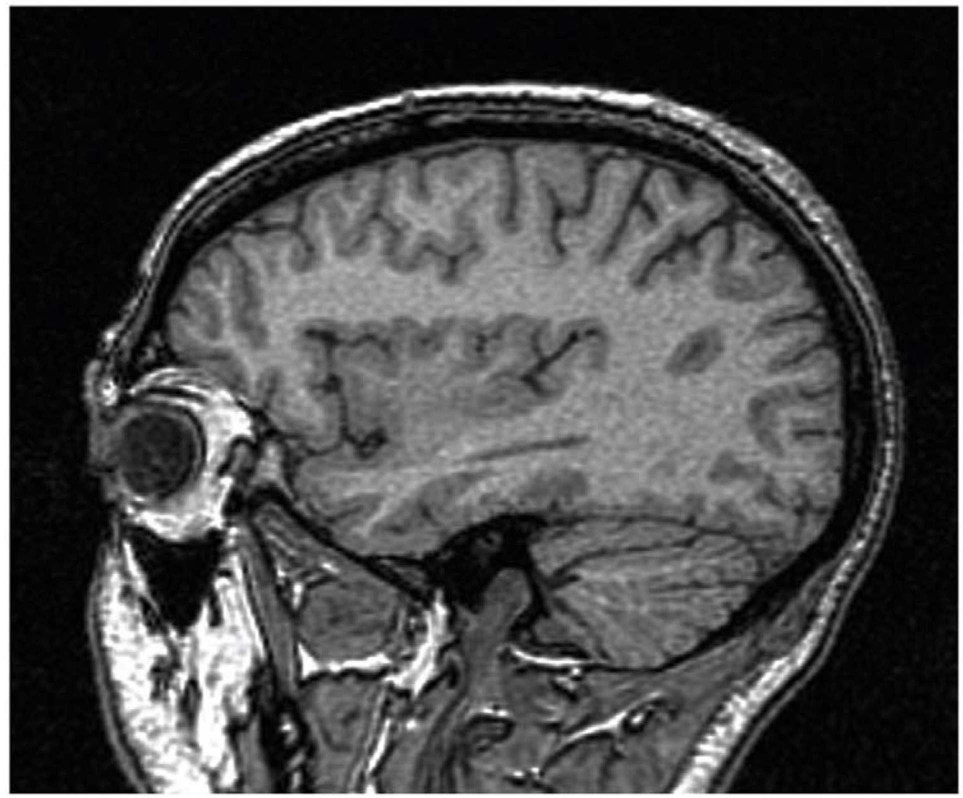
脑科学是最具挑战性的前沿科学领域



爱因斯坦大脑切片

脑核磁成像技术（MRI）

(B)



NEUROSCIENCE, Fourth Edition, Atlas, Plate 4 (Part 2)

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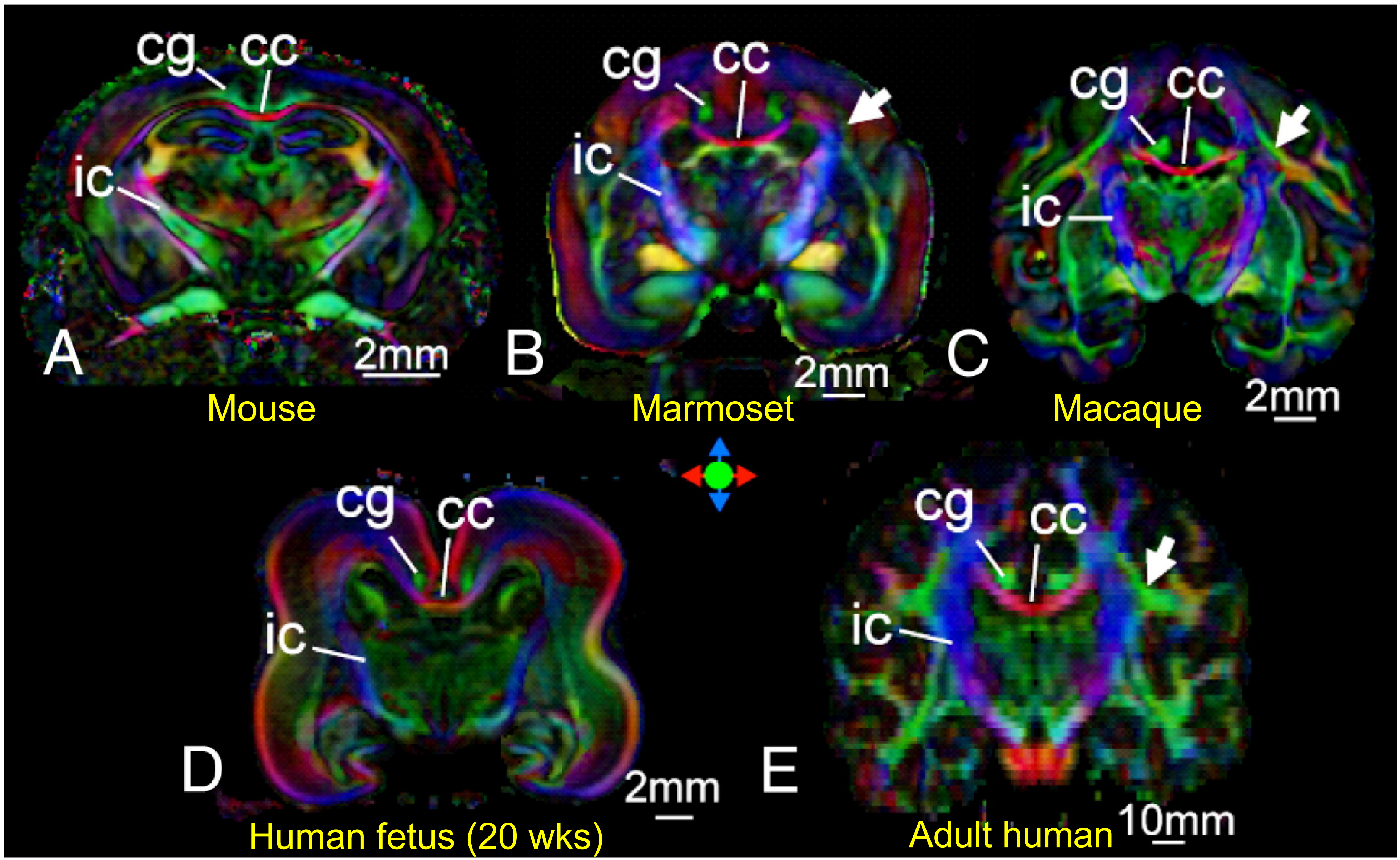
Fiber Tracks Revealed by Diffusion Tensor MRI



Zhang et al. (*NeuroImage* 2007)

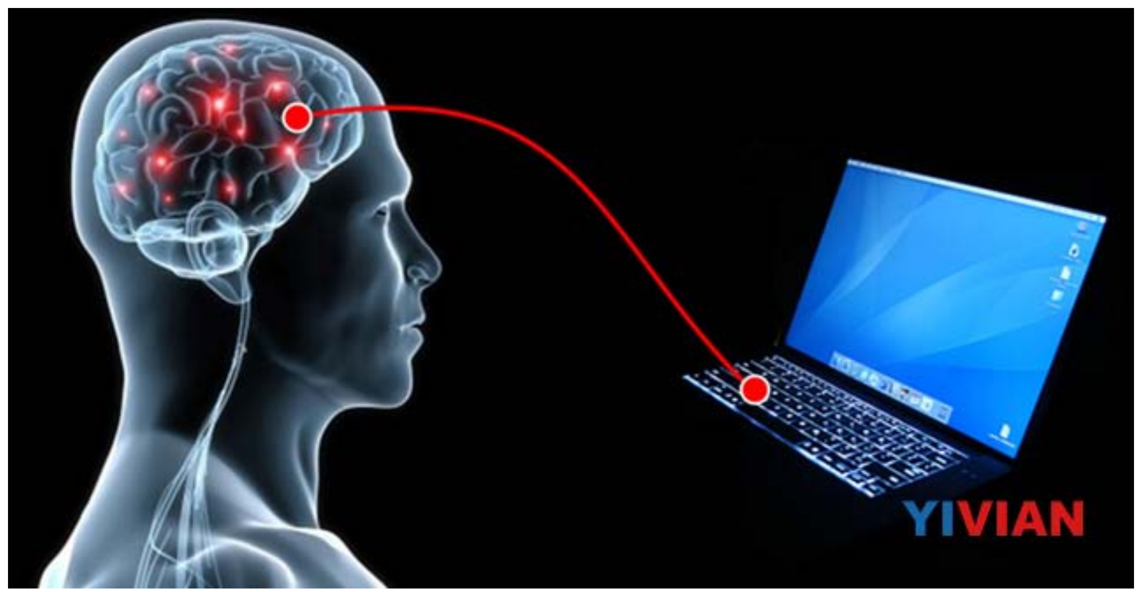
不仅仅是科学家对大脑的奥秘有兴趣

NEURALINK

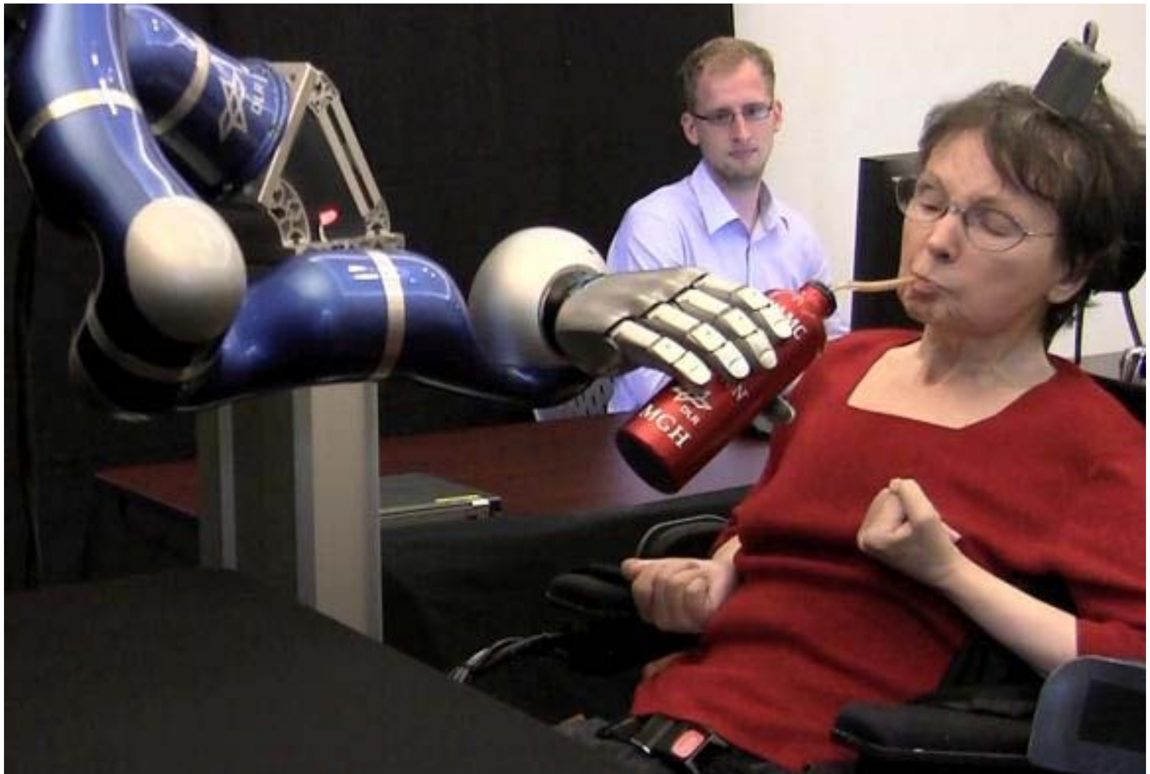
Neuralink wants to wire your brain to the internet

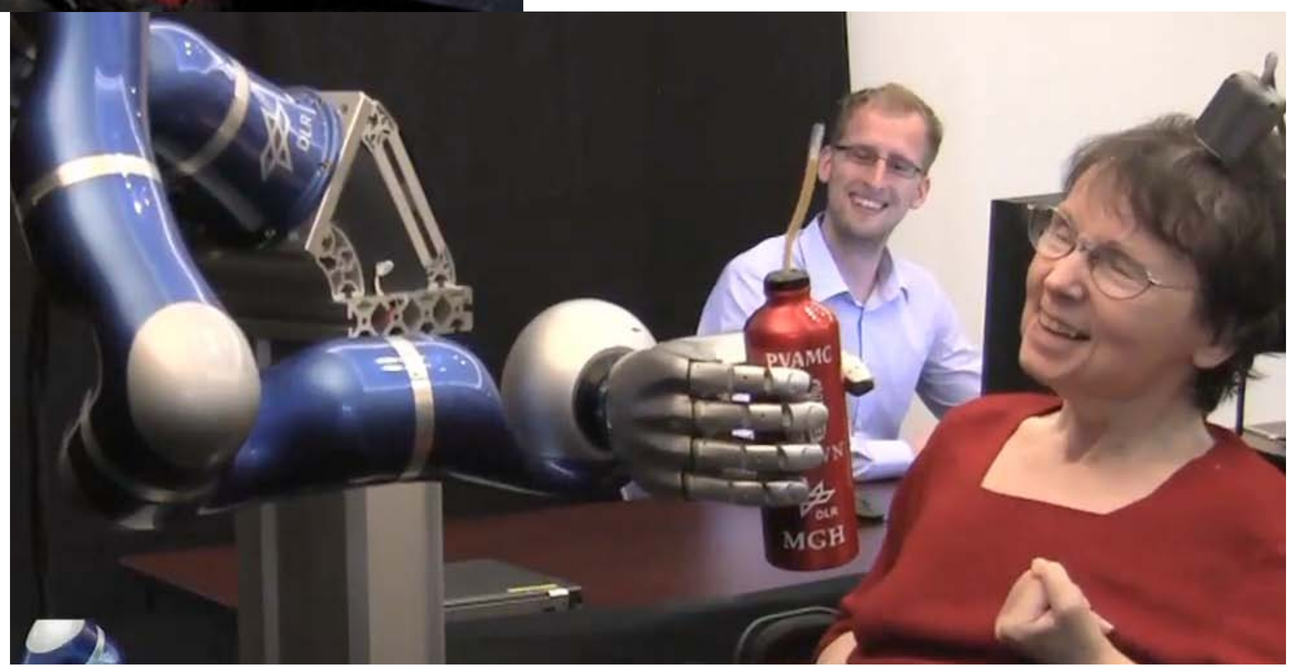
脑机接口

Brain-Machine Interface (BCI)



《经济学人》:脑机接口技术是下一个风口


MGH


PVAMC
MGH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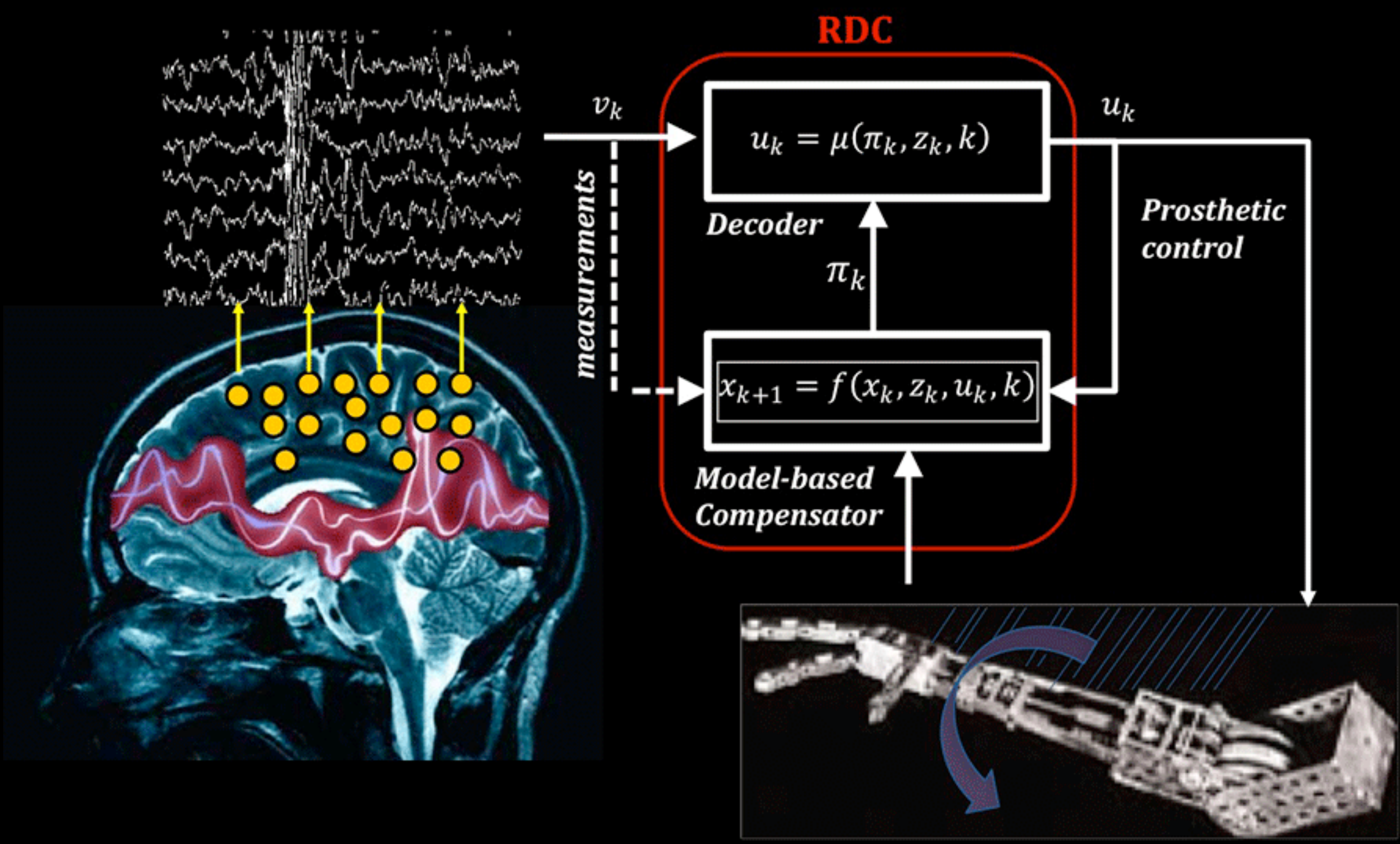
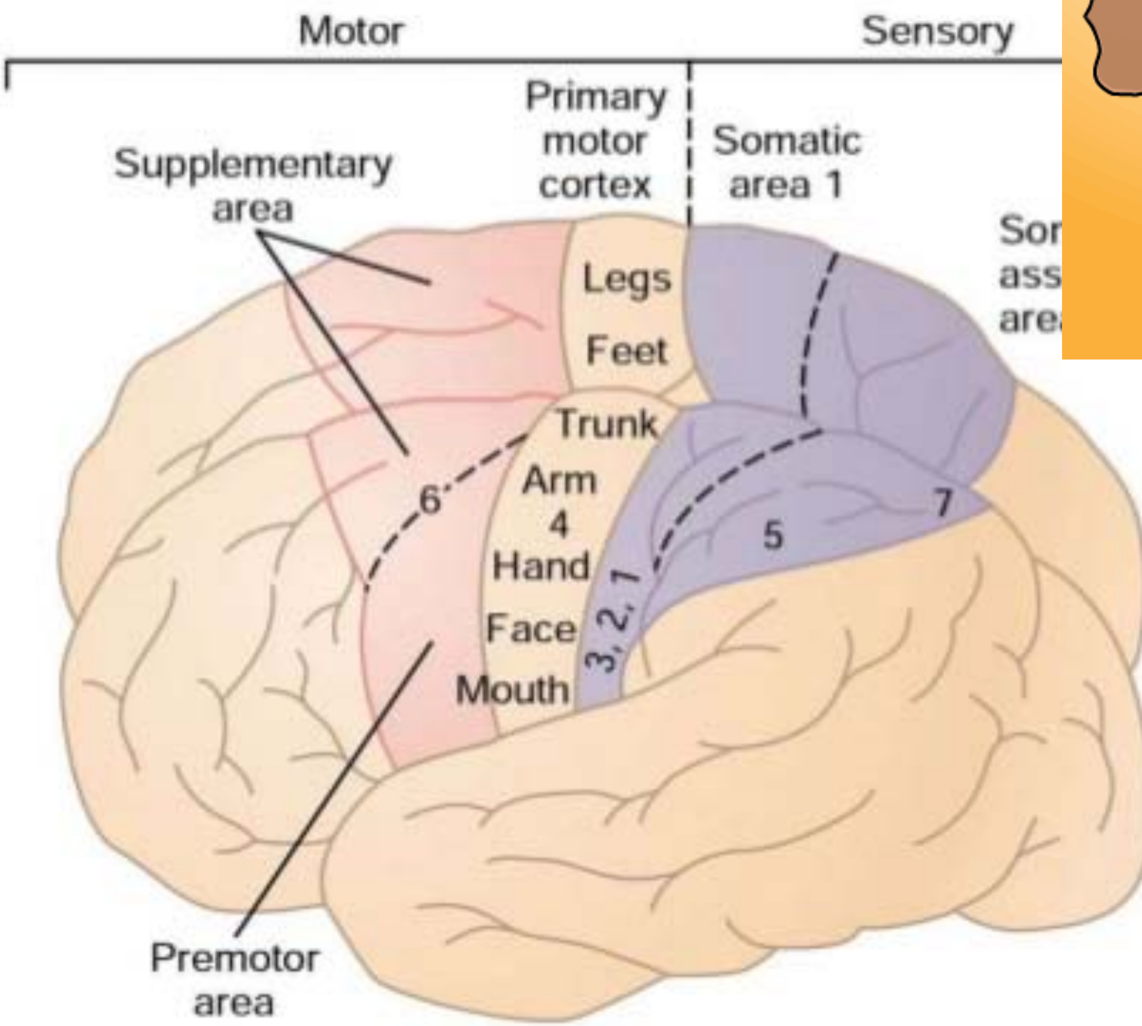
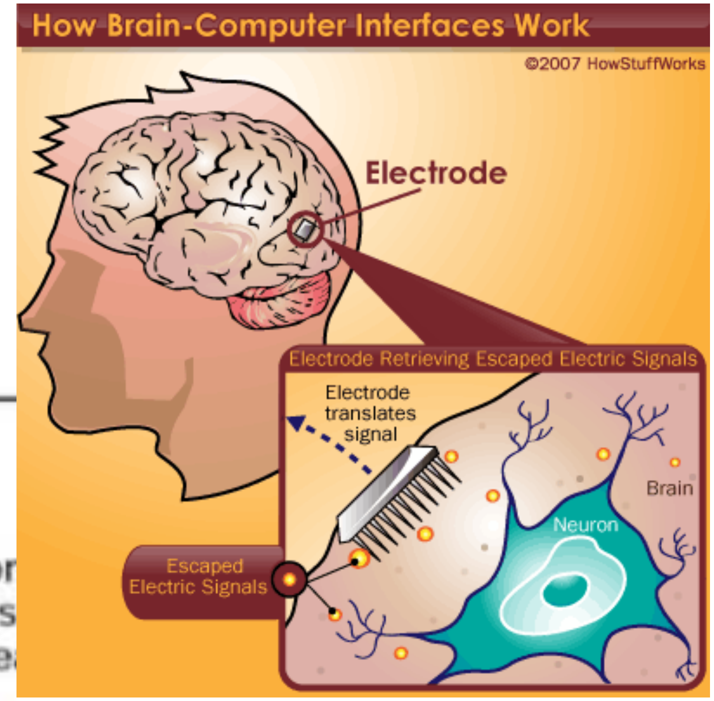
脑机接口的工作原理

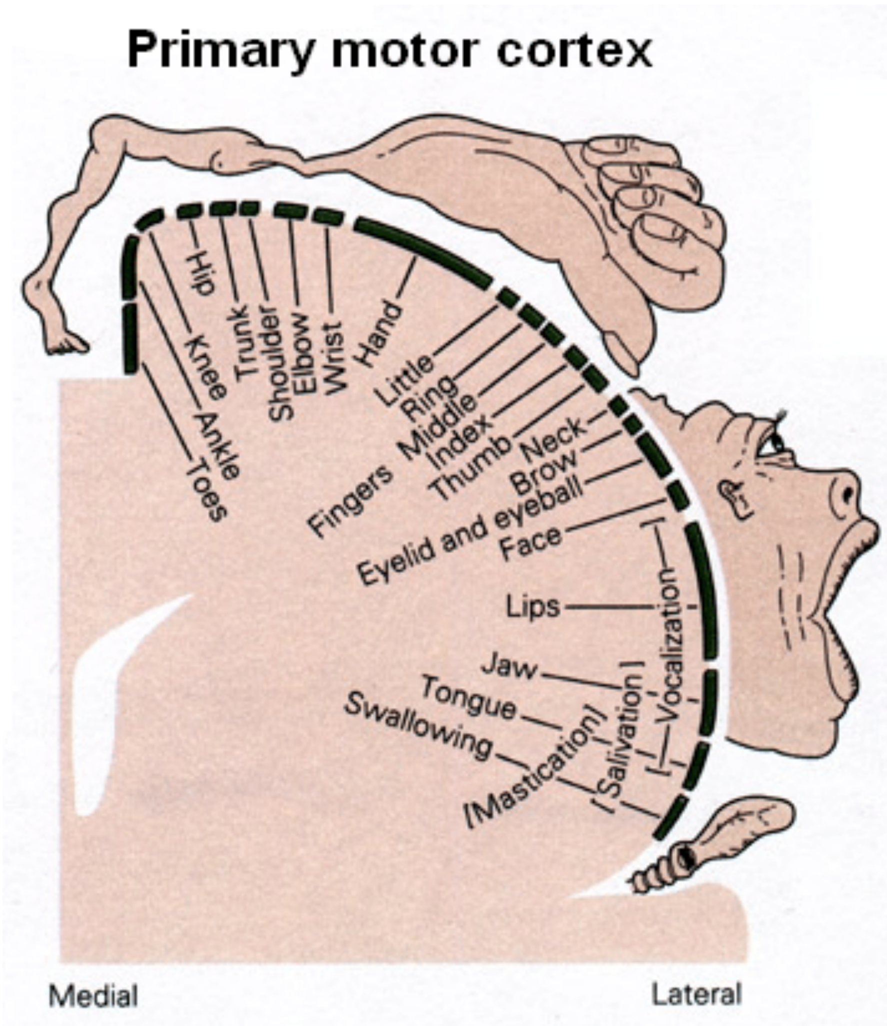


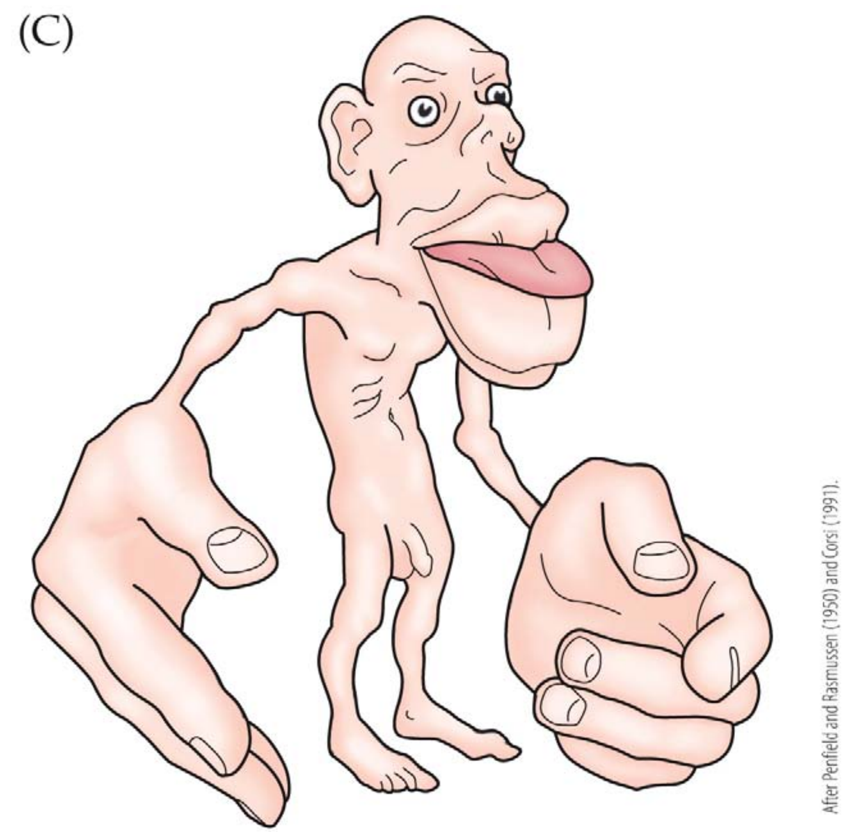
RDC
v_k
measurements
$u_k = \mu(\pi_k, z_k, k)$
Decoder
π_k
$x_{k+1} = f(x_k, z_k, u_k, k)$
Model-based Compensator
u_k
Prosthetic control


Motor
Sensory
Supplementary area
Primary motor cortex
Somatic area 1
Legs
Feet
Trunk
Arm
4
Hand
Face
Mouth
6
3, 2, 1
5
7
Premotor area


How Brain-Computer Interfaces Work
©2007 HowStuffWorks
Electrode
Electrode Retrieving Escaped Electric Signals
Electrode translates signal
Brain
Neuron
Escaped Electric Signals

Motor Homunculus





NEUROSCIENCE 6e, Figure 9.11 (Part 3)
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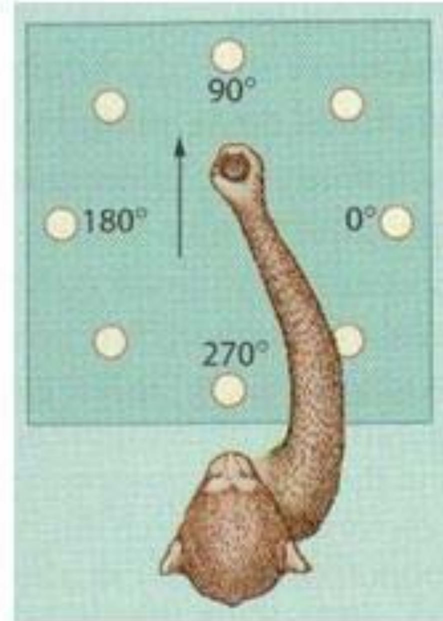
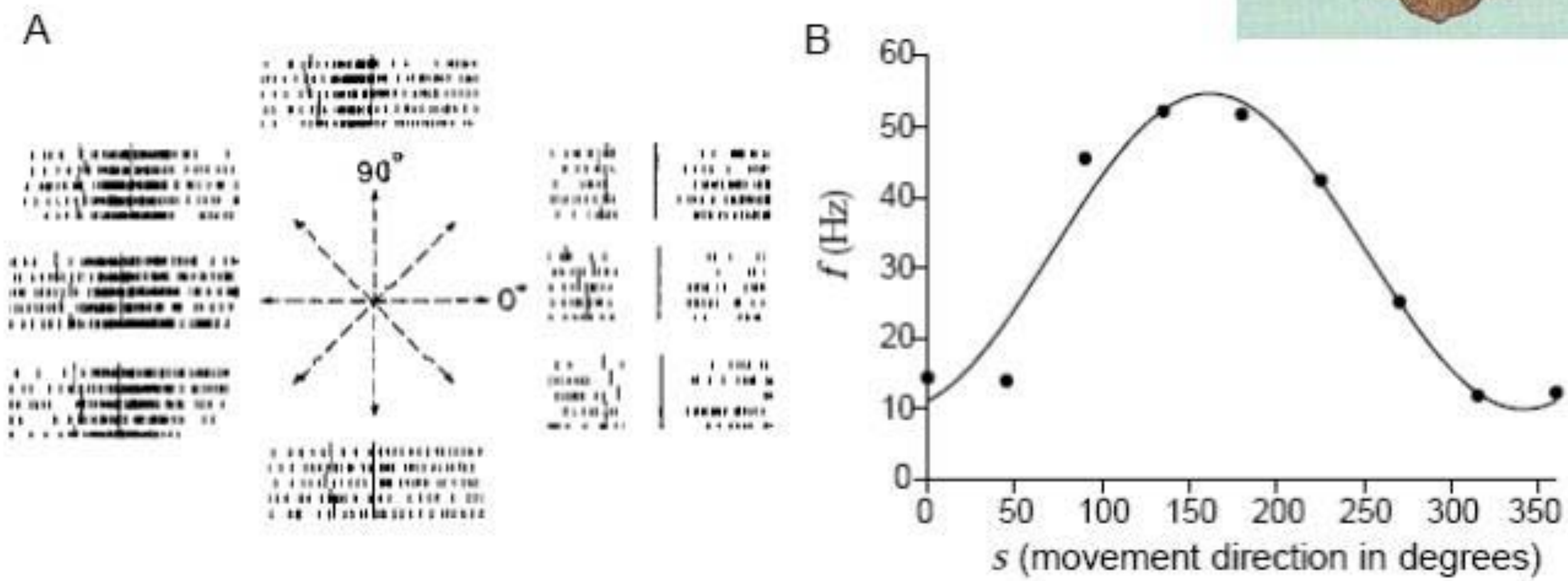
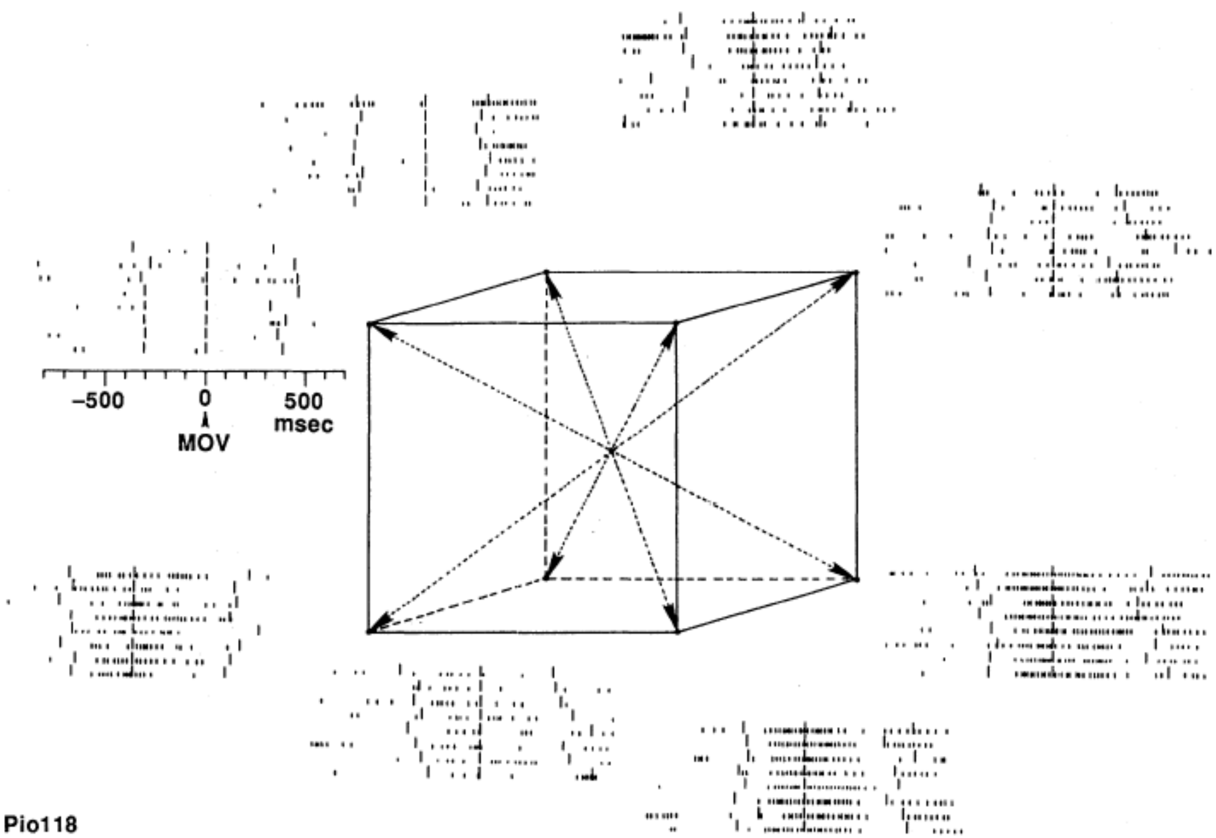
Decoding arm movement direction in M1 cortex of the monkey by population vector method

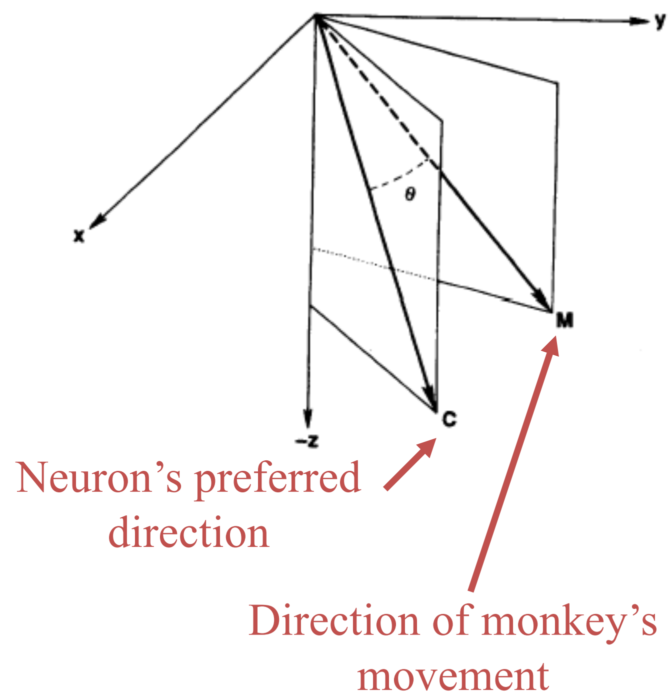




Recordings from the primary motor cortex of a monkey performing an arm reaching task

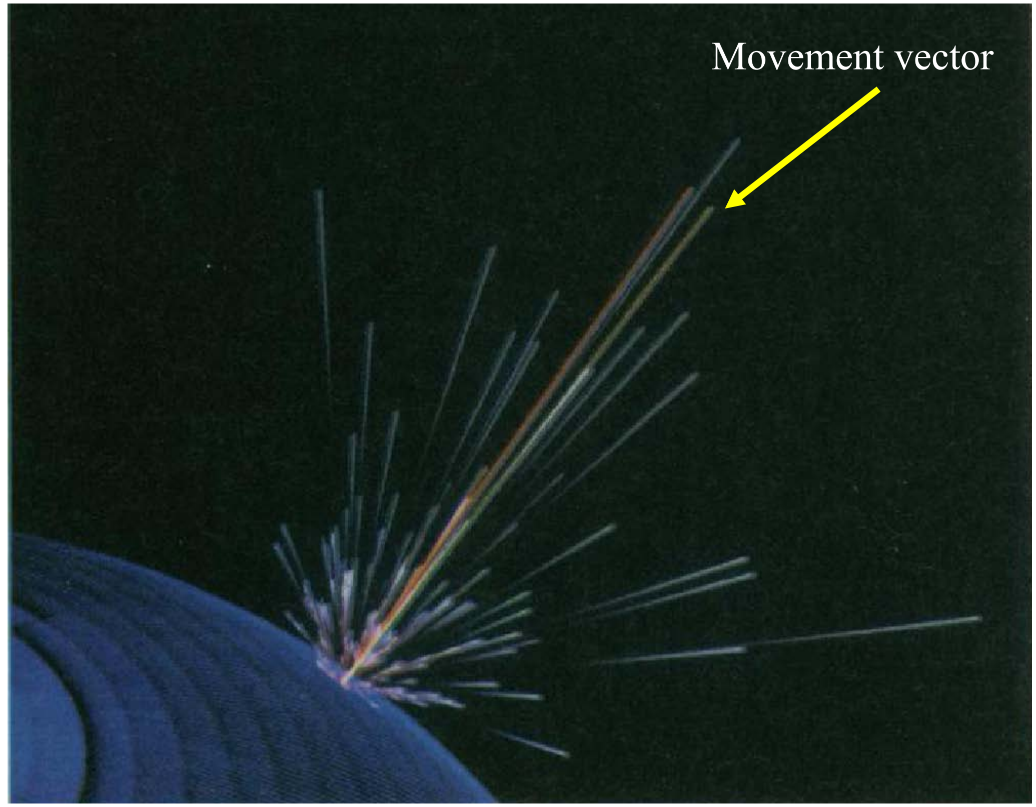
Directional tuning of individual neurons in motor cortex





Georgopoulos et al. (Science, 1986)

Population vector predicts monkey's movement



Assumptions:

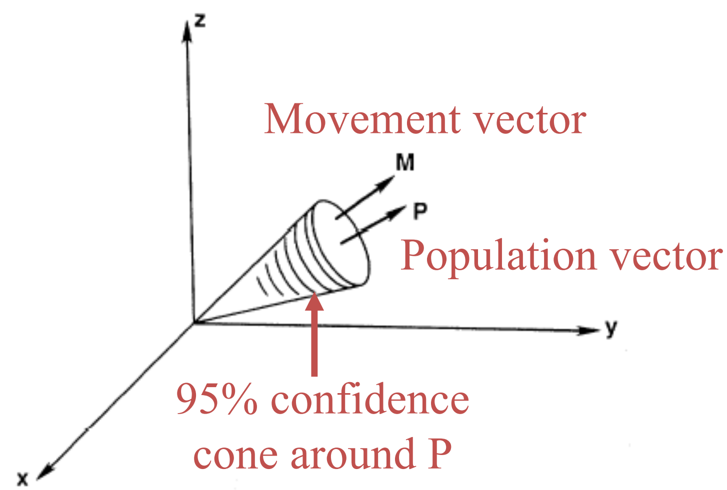
1) Each neuron makes a vectorial contribution along its "preferred direction" **C.**
2) The magnitude of the contribution of each neuron is a function of the movement direction, equal to mean firing rate for movement in direction **M.**

i-th neuron's contribution is:

$$\mathbf{N}_i\,(\mathbf{M}) = w_i\,(\mathbf{M})\;\mathbf{C}_i$$

Population vector of 224 neurons is:

$$\mathbf{P}(\mathbf{M}) = \sum_{i=1}^{224} \mathbf{N}_i(\mathbf{M})$$



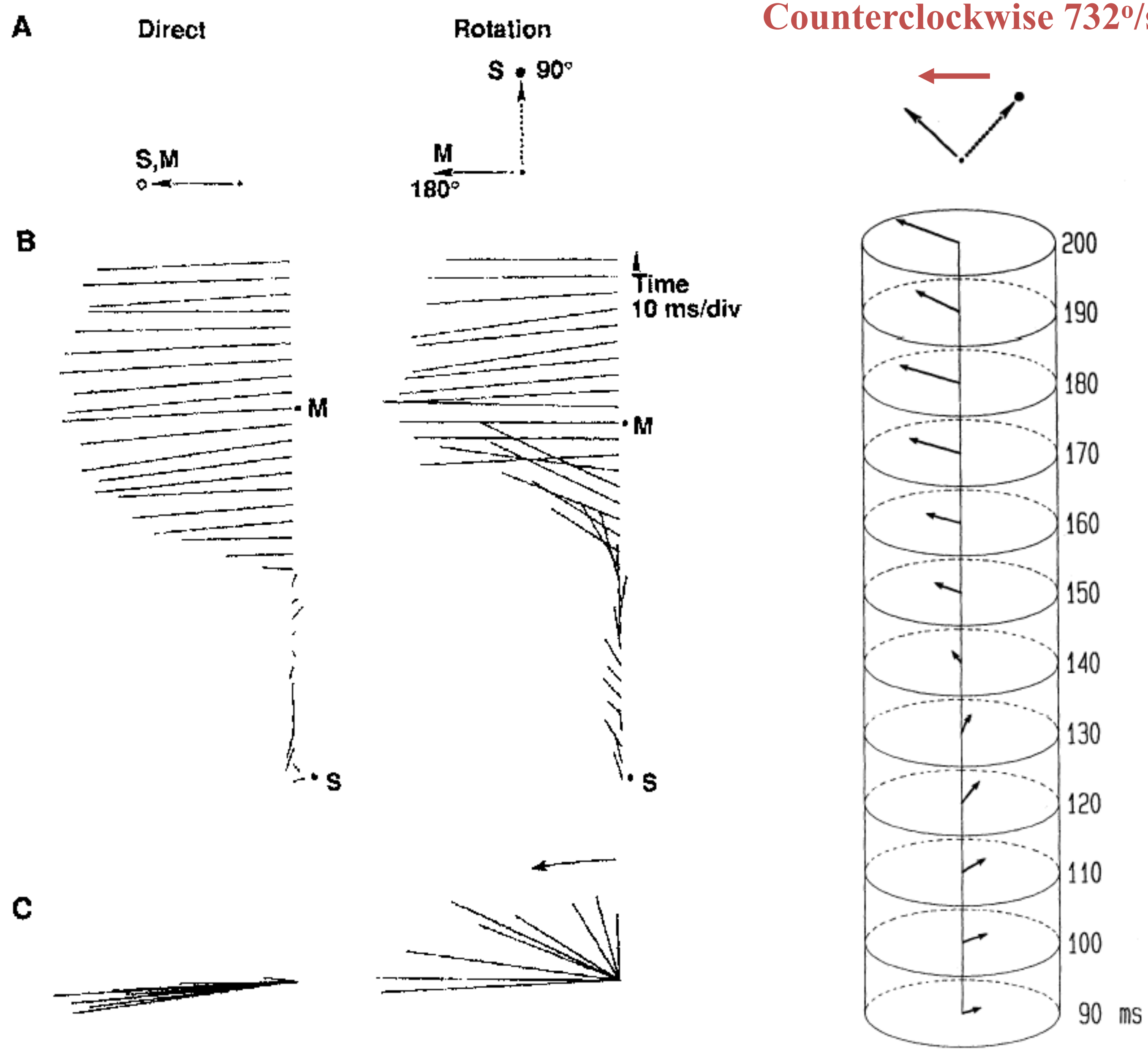
Georgopoulos et al. (Science, 1986)

Mental rotation analyzed by Population Vector

S (stimulus)

M (movement)

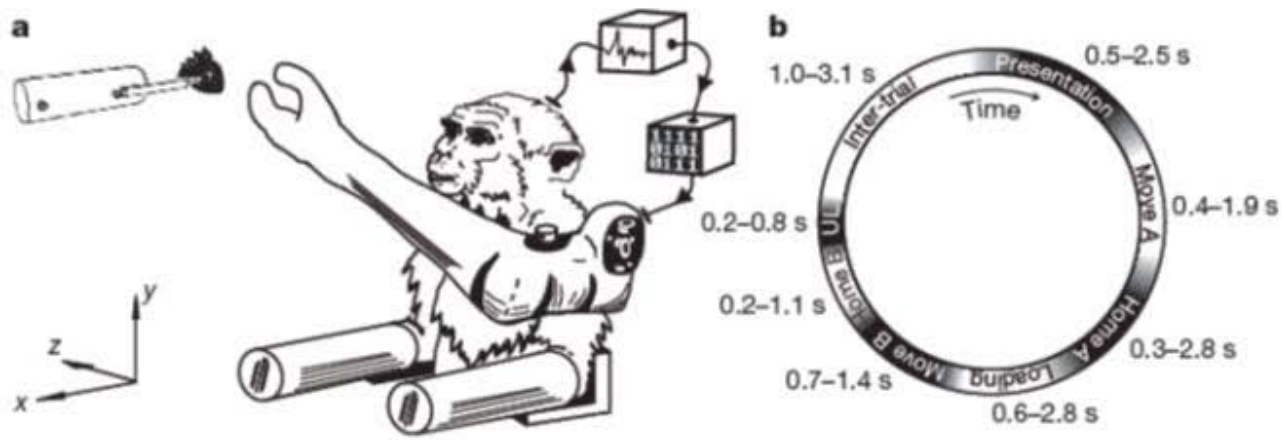
Counterclockwise 732°/sec



Georgopoulos et al. (Science, 1989)

Cortical control of a prosthetic arm for self-feeding

Meel Velliste[1], Sagi Perel[2,3], M. Chance Spalding[2,3], Andrew S. Whitford[2,3] & Andrew B. Schwartz[1–6]

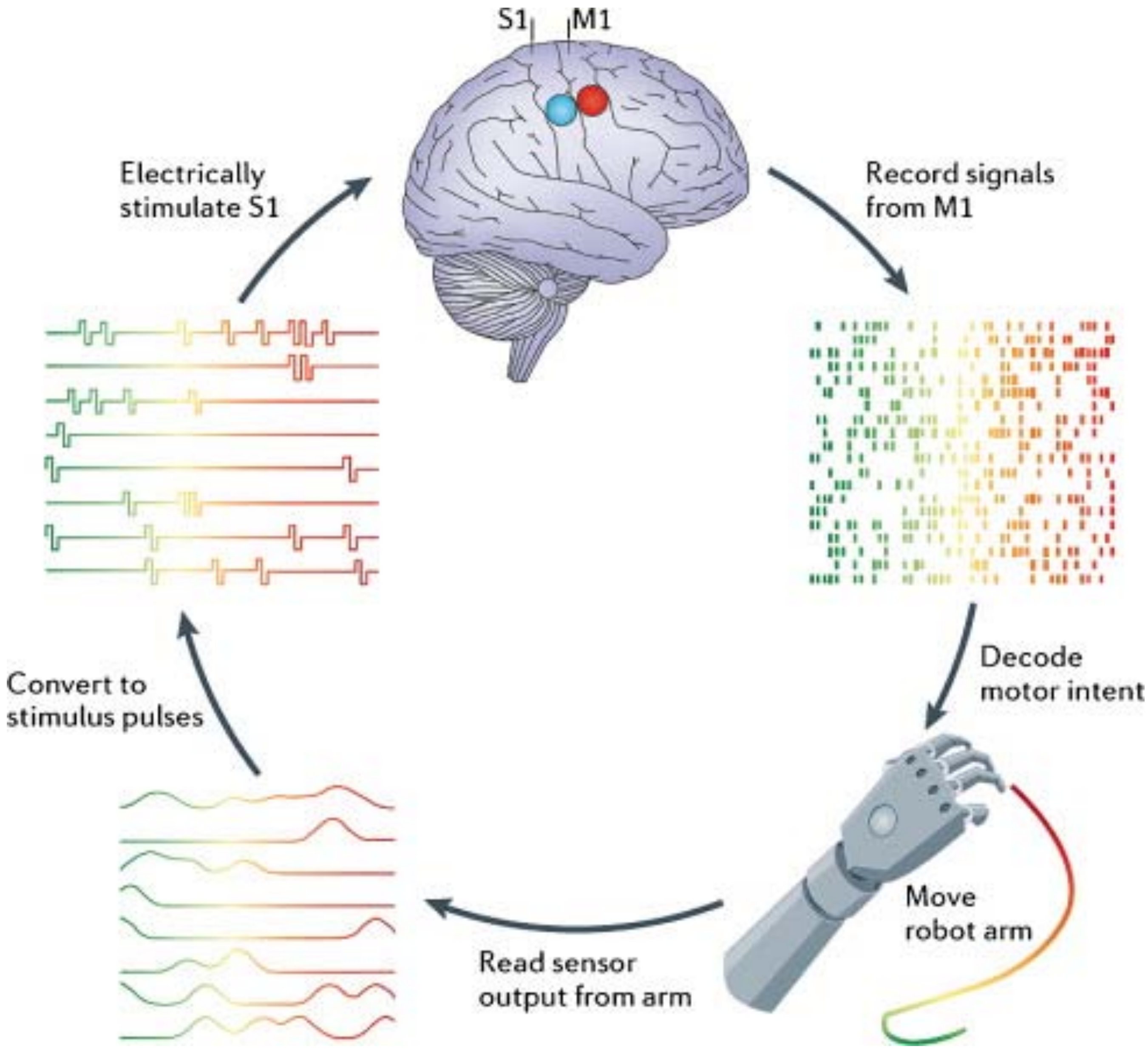


1. The monkeys are trained to operate the arm for a self-feeding task using a joystick

2. Their own arms were then restrained and the prosthetic arm was controlled with populations of motor cortex single- and multi-unit spiking activity.

Velliste et al., 2008

双向脑机接口


S1
M1
Record signals
from M1
Decode
motor intent
Move
robot arm
Read sensor
output from arm
Convert to
stimulus pulses
Electrically
stimulate S1

第二个例子

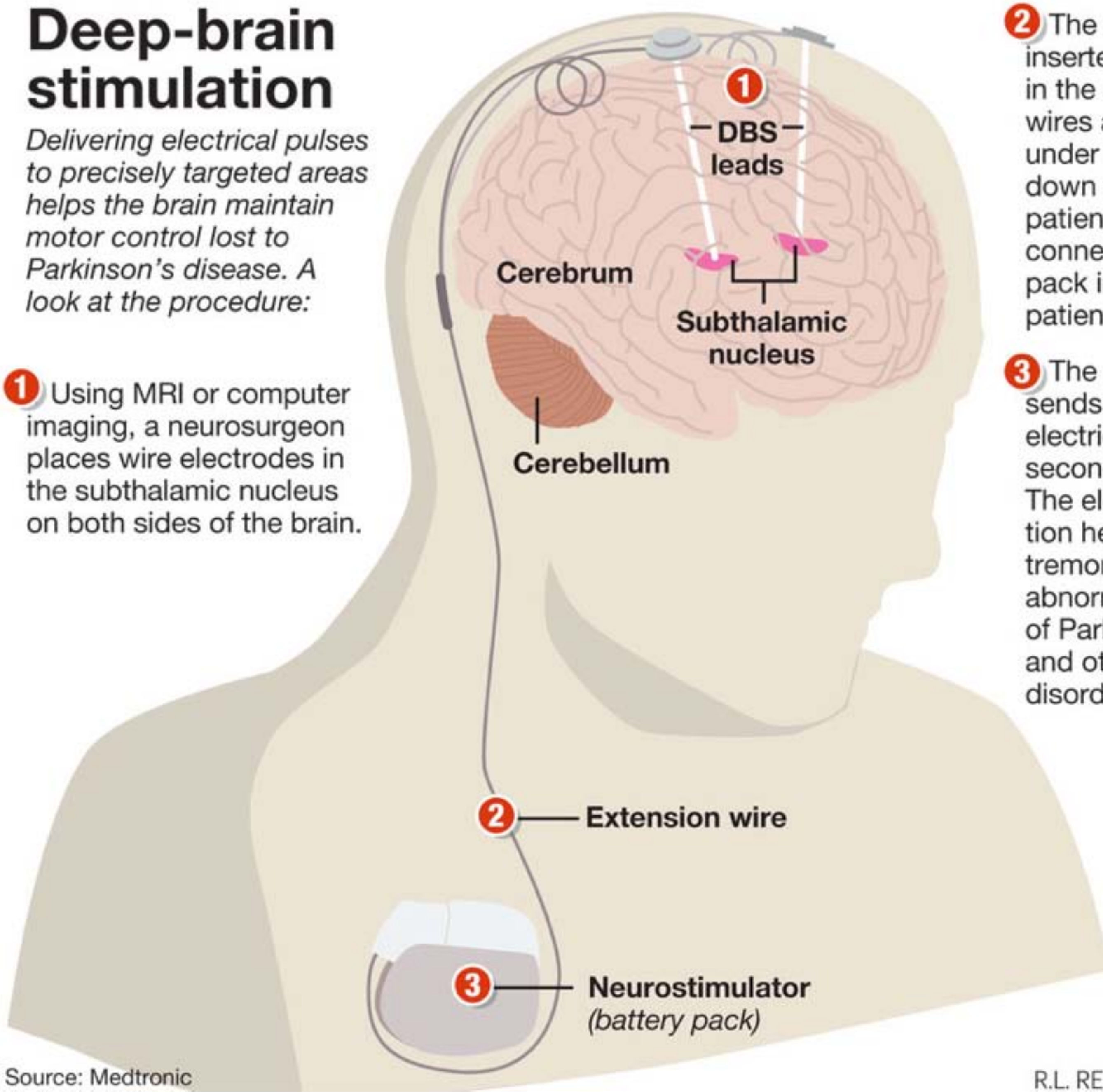
Deep Brain Stimulation (DBS)

“脑起搏器”

Deep-brain stimulation

Delivering electrical pulses to precisely targeted areas helps the brain maintain motor control lost to Parkinson's disease. A look at the procedure:

1. Using MRI or computer imaging, a neurosurgeon places wire electrodes in the subthalamic nucleus on both sides of the brain.



2. The leads are inserted through holes in the skull. Extension wires are threaded under the skin and down the side of the patient's head, then connected to a battery pack implanted in the patient's chest.

3. The battery pack sends more than 100 electrical pulses a second to the brain. The electrical stimulation helps control the tremors and other abnormal movements of Parkinson's disease and other movement disorders.

Source: Medtronic

R.L. REBACH/STAFF ARTIST


Medtronic
ACTIVA RC

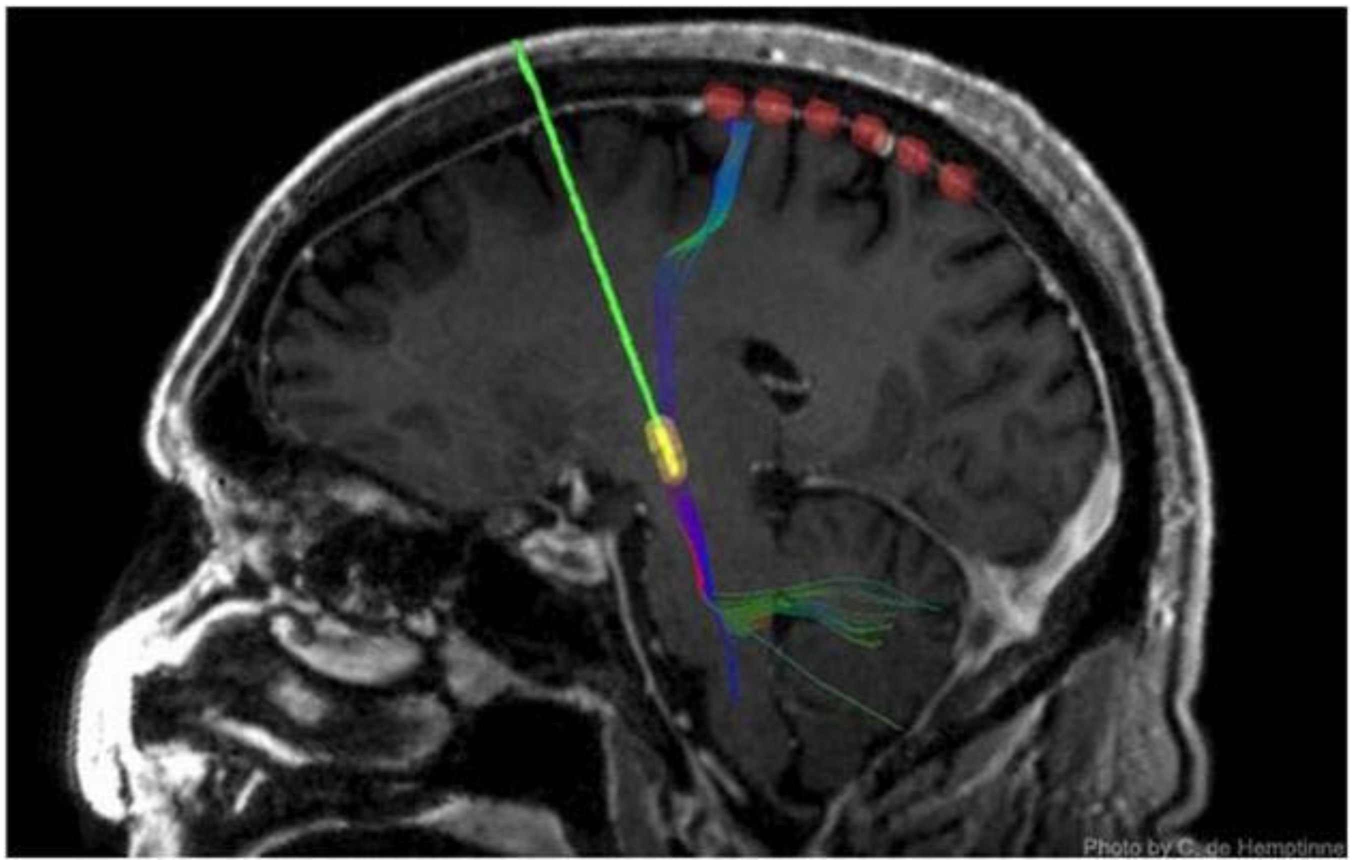


Photo by C. de Hemptinne

第三个例子

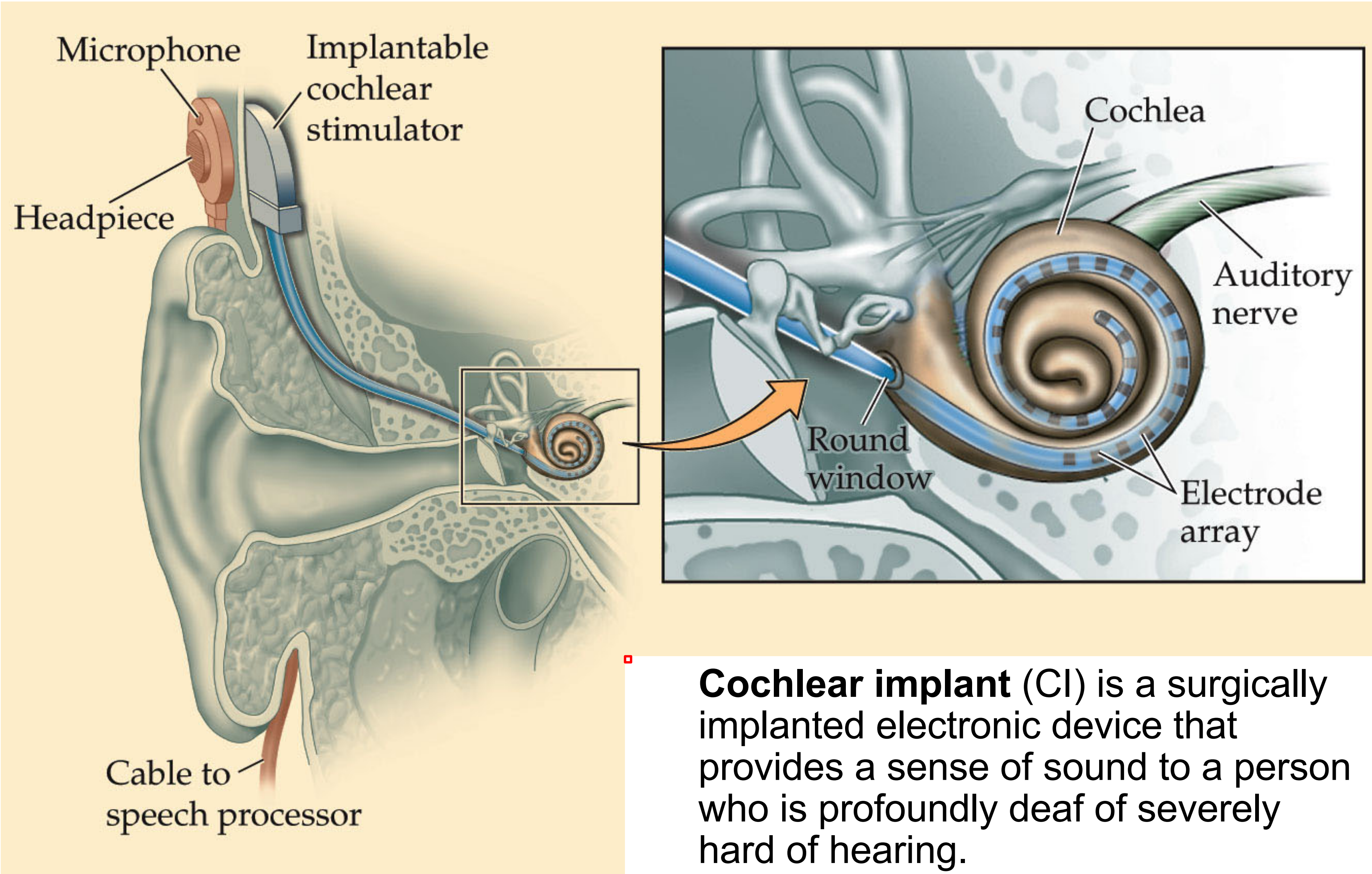
Cochlear Implant

电子耳蜗

Deafness and Hearing Loss

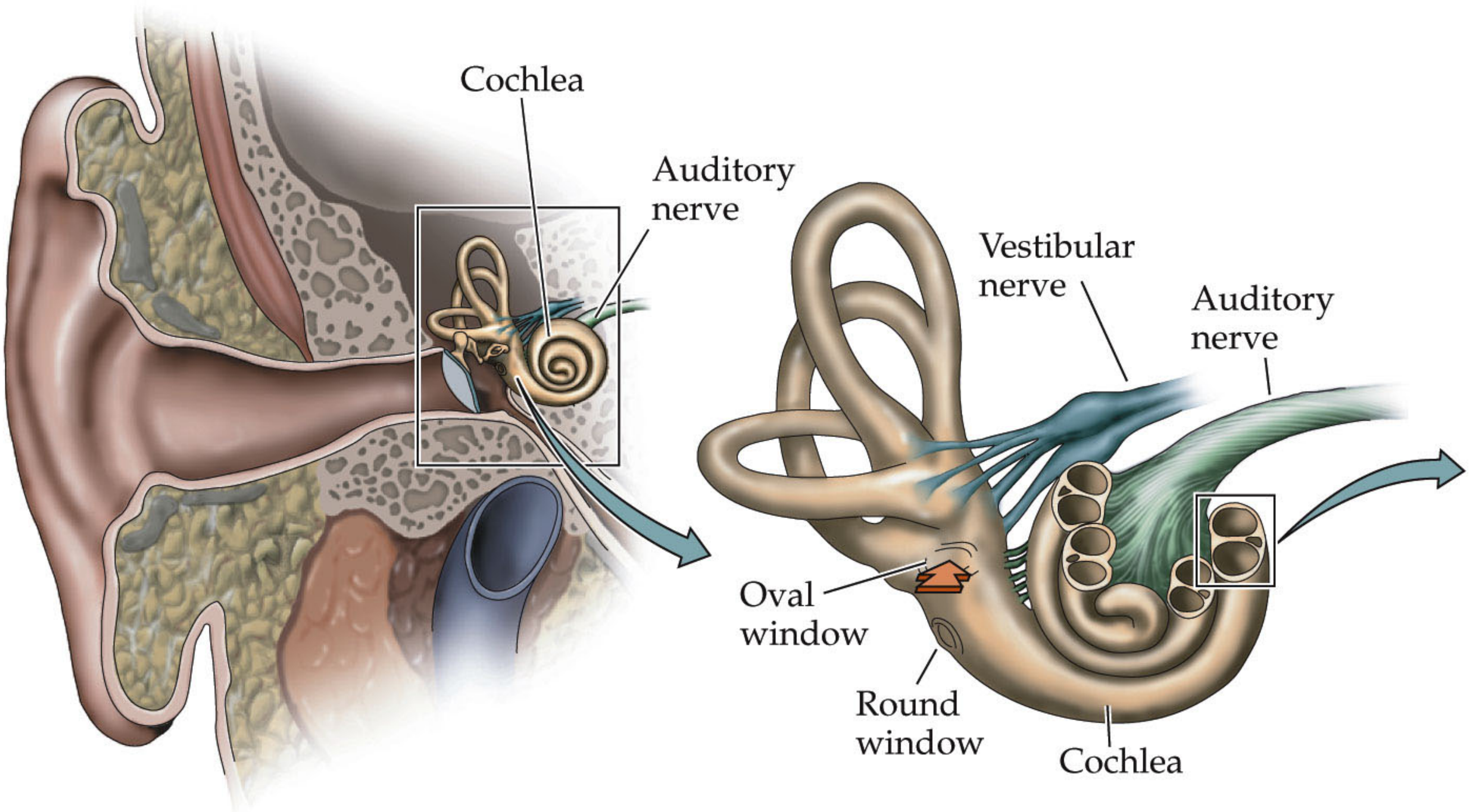
- Hearing loss affects approximately 17 in 1,000 children under age 18 in the United States.
- About 2 to 3 out of every 1,000 children in the U.S. are born deaf or hard-of-hearing due to genetic dispositions or diseases.
- A profound consequence of the deafness in childhood is the inability to communicate with speech, which often results in delayed language learning and intellectual development, and consequently problems at school.
- Cochlear implant (CI) is the most successful neural prosthesis device that has restored hearing in a large number of deaf children and adults

What is cochlear implant?



Cochlear implant (CI) is a surgically implanted electronic device that provides a sense of sound to a person who is profoundly deaf of severely hard of hearing.

How does cochlear implant work?



NEUROSCIENCE, Fourth Edition, Figure 13.4 (Part 1)

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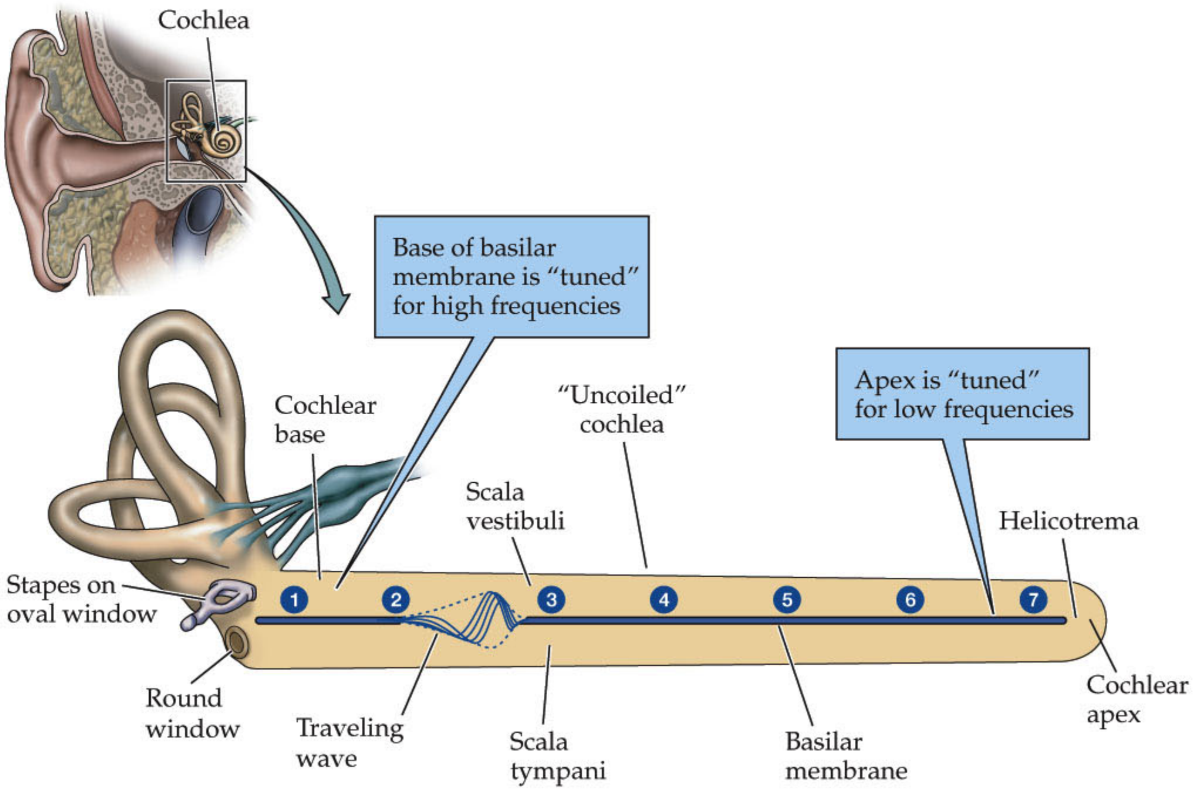
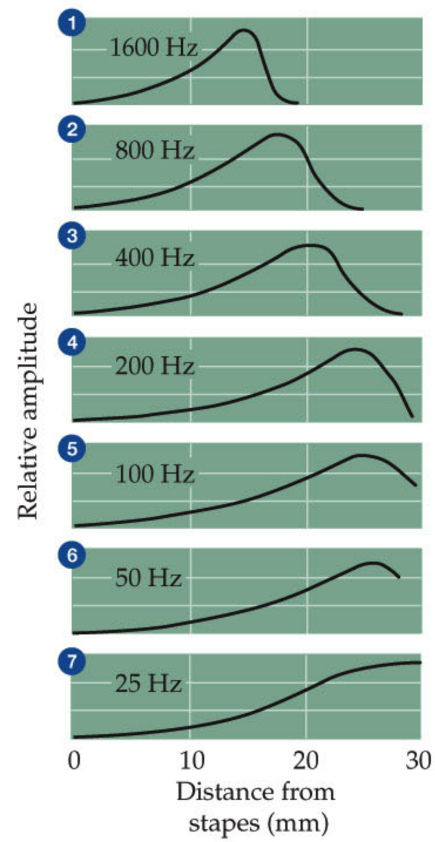
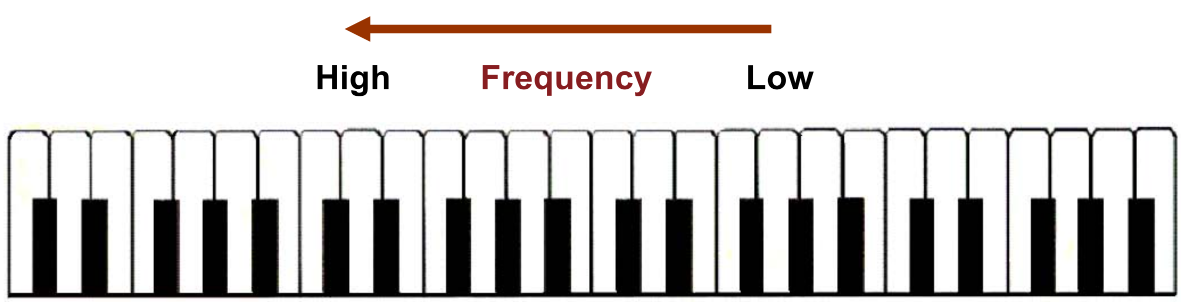
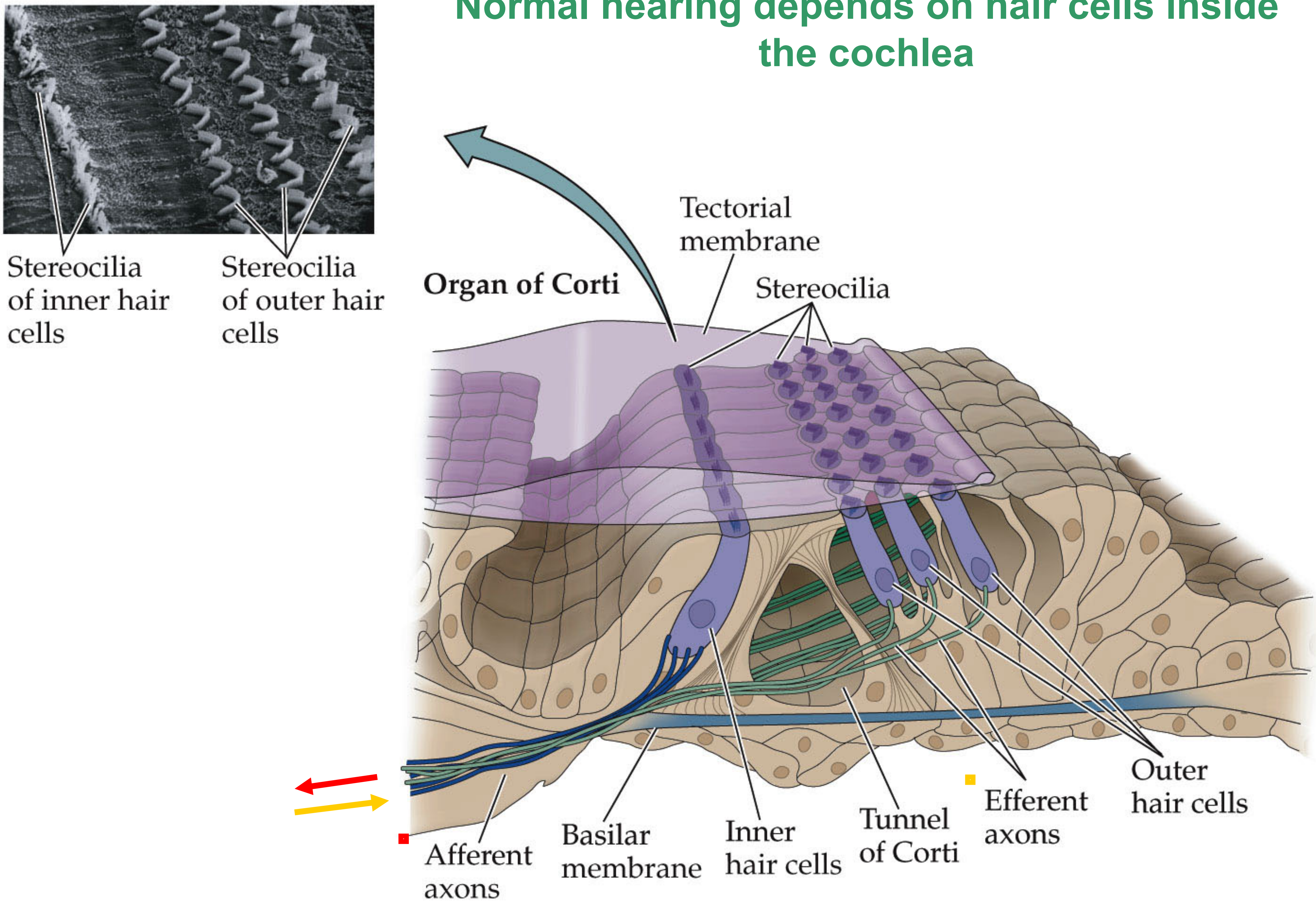
Cochlea performs frequency decomposition







Normal hearing depends on hair cells inside the cochlea



NEUROSCIENCE, Fourth Edition, Figure 13.4 (Part 3)

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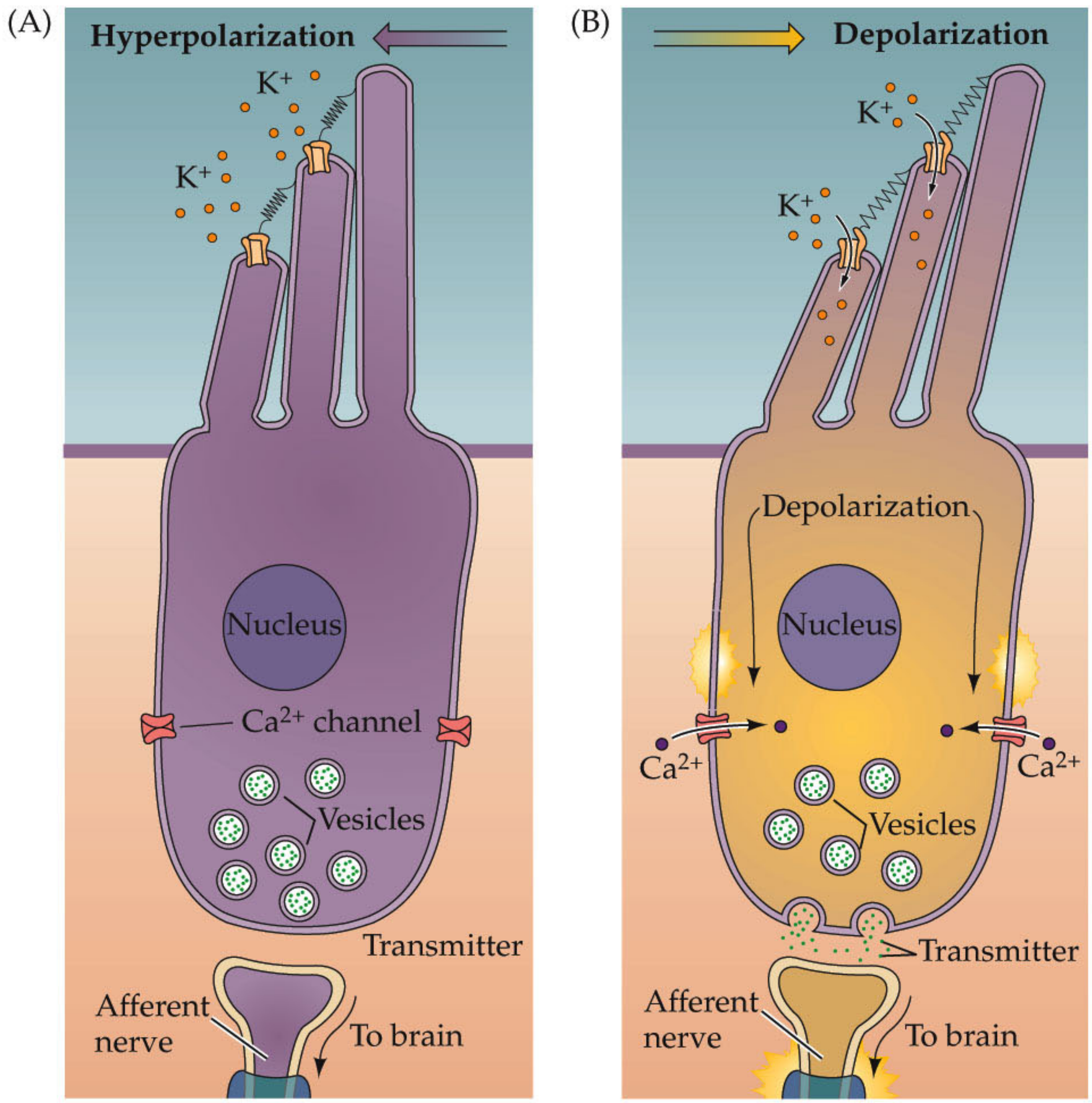
Mechanoelectrical transduction mediated by hair cells



NEUROSCIENCE, Fourth Edition, Figure 13.8

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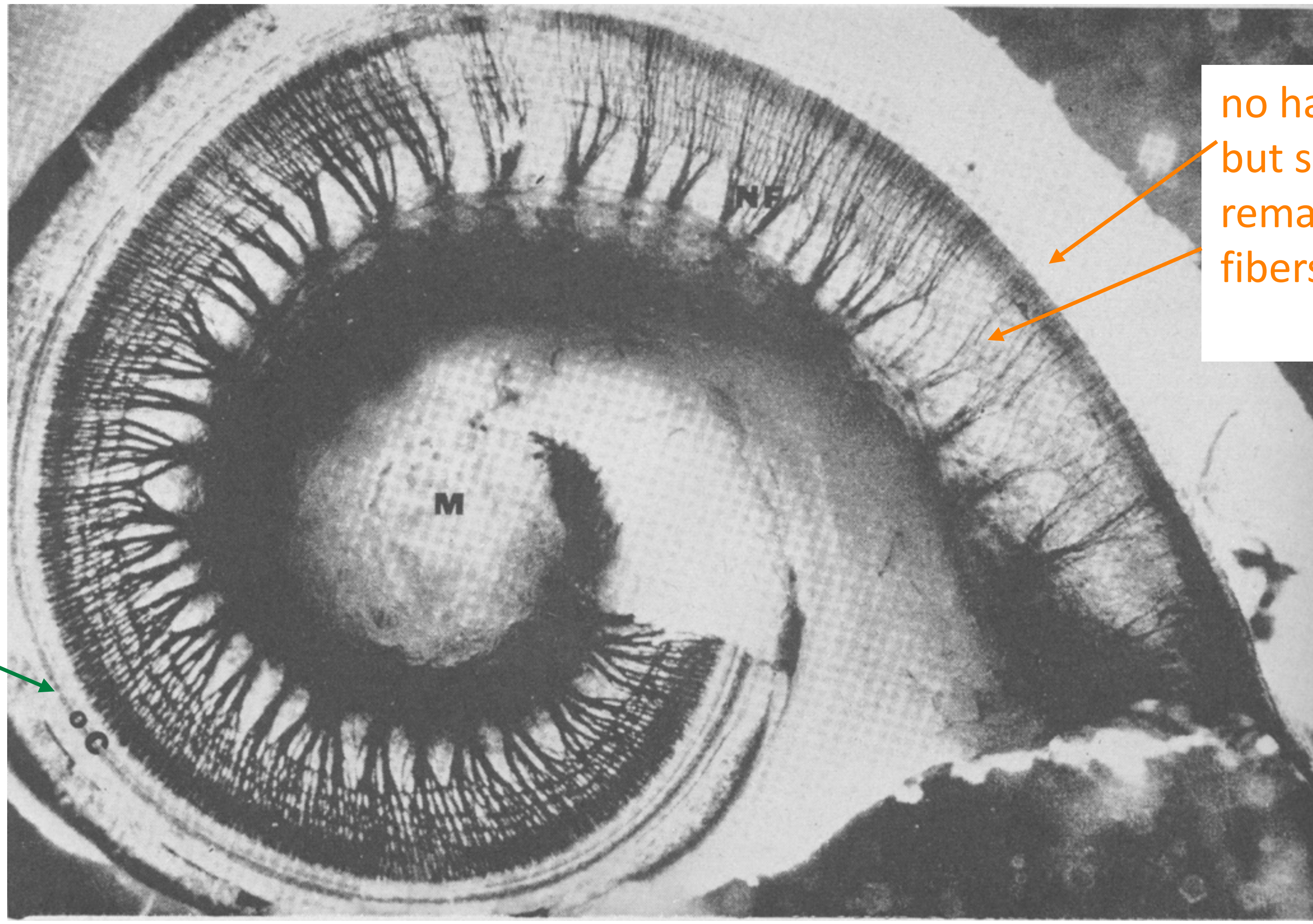
Fortunately, the auditory nerve fibers often survive the loss of hair cells (typically 10-70% of fibers survive)

The idea of a cochlear implant is to electrically stimulate these remaining fibers

remaining hair cells

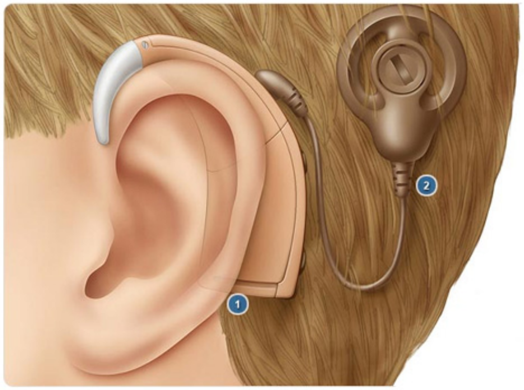


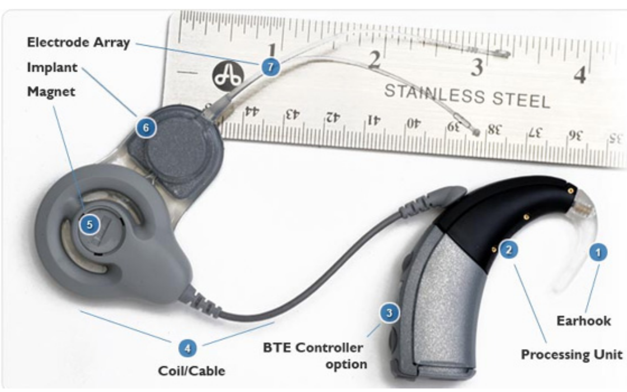
no hair cells, but some remaining fibers

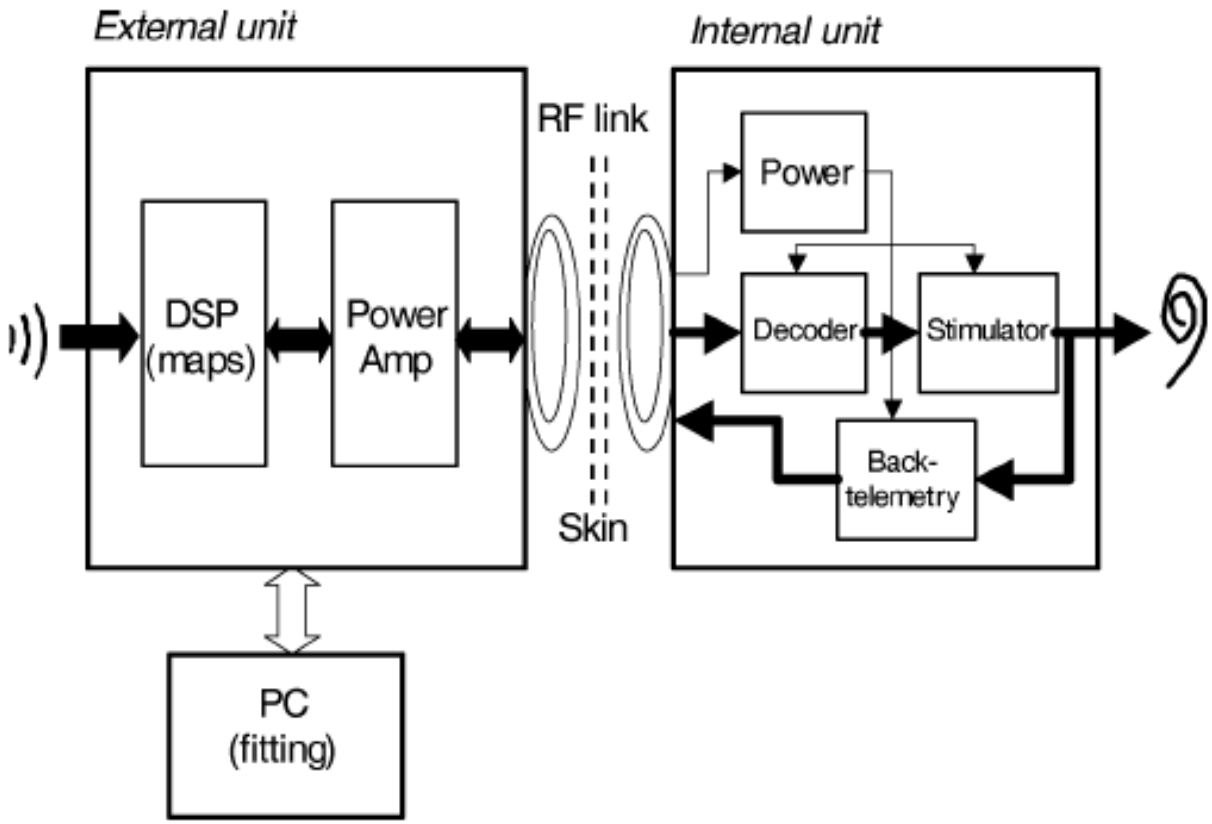
The basal turn of the cochlea of a 93 year old man with considerable hair-cell loss

Bredberg, 1968

Designs of Cochlear Implant Device

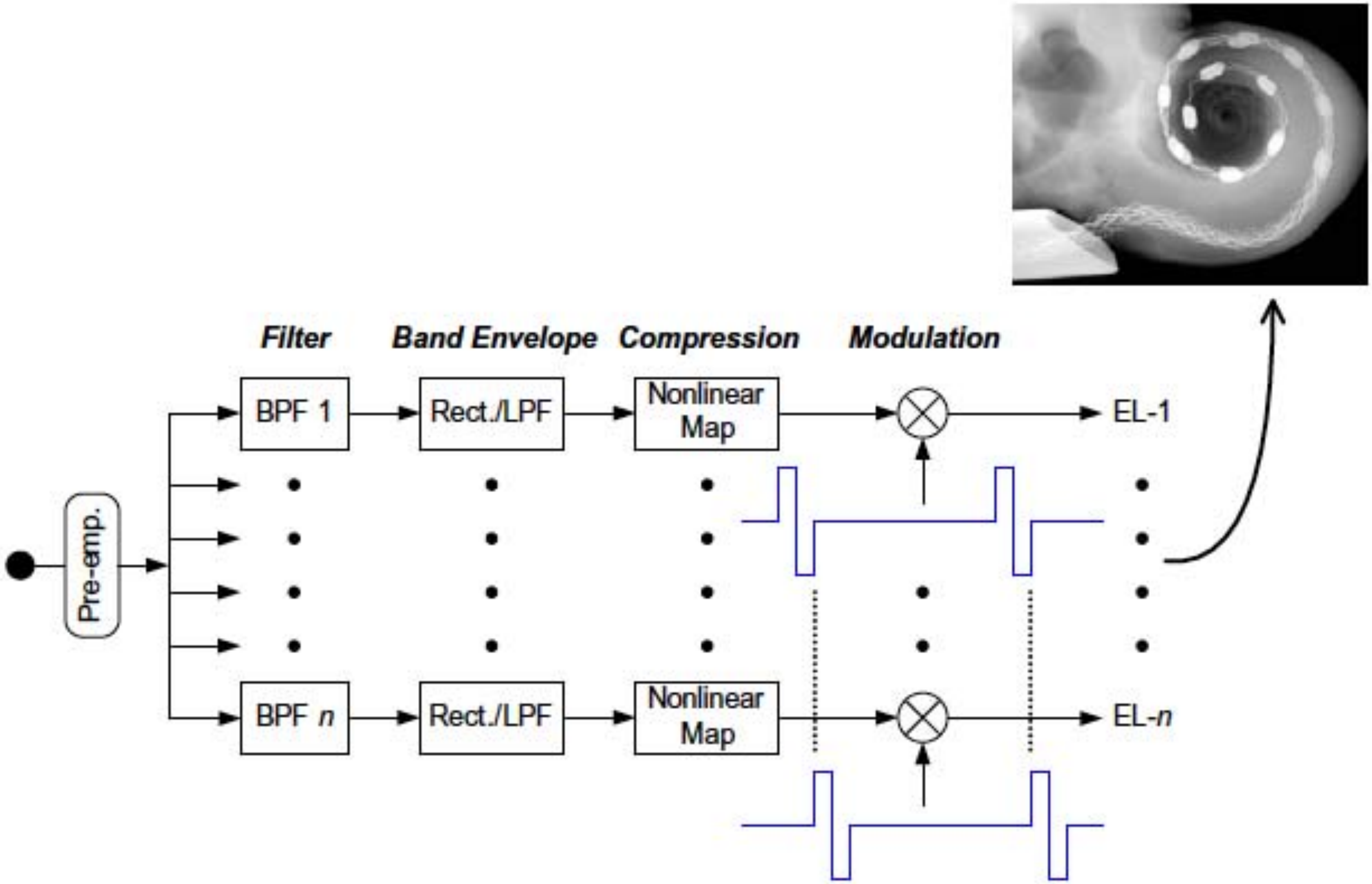






Zeng (IEEE Rev BME, 2008)

Multi-channel Cochlear Implant Signal Processing Strategy

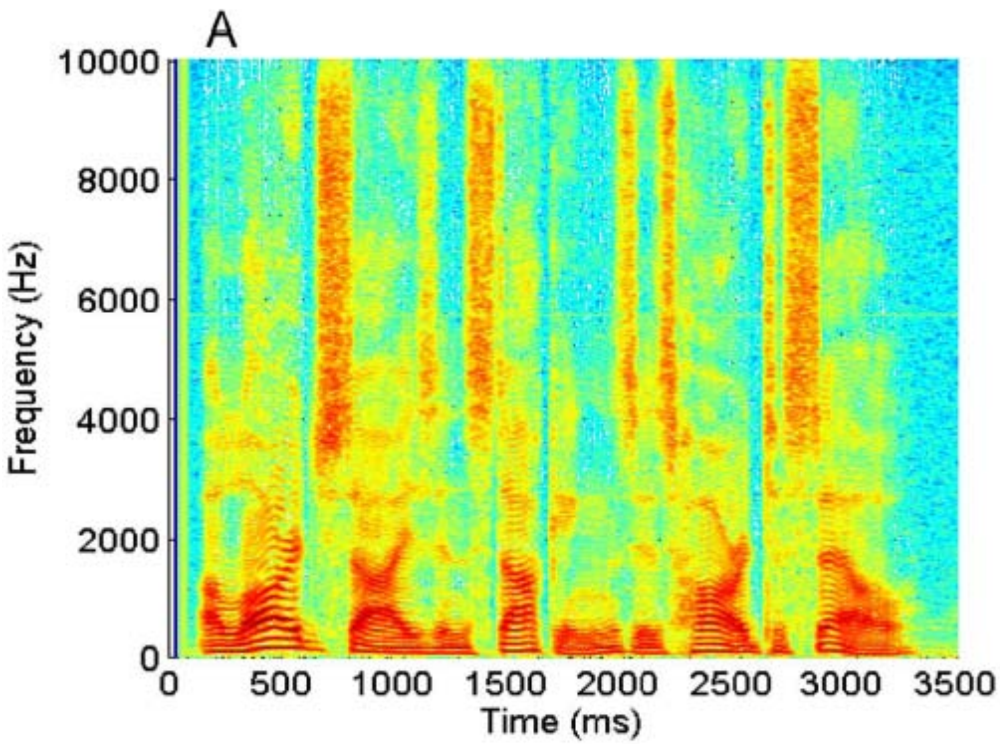


Wilson et al. (Nature 1991)

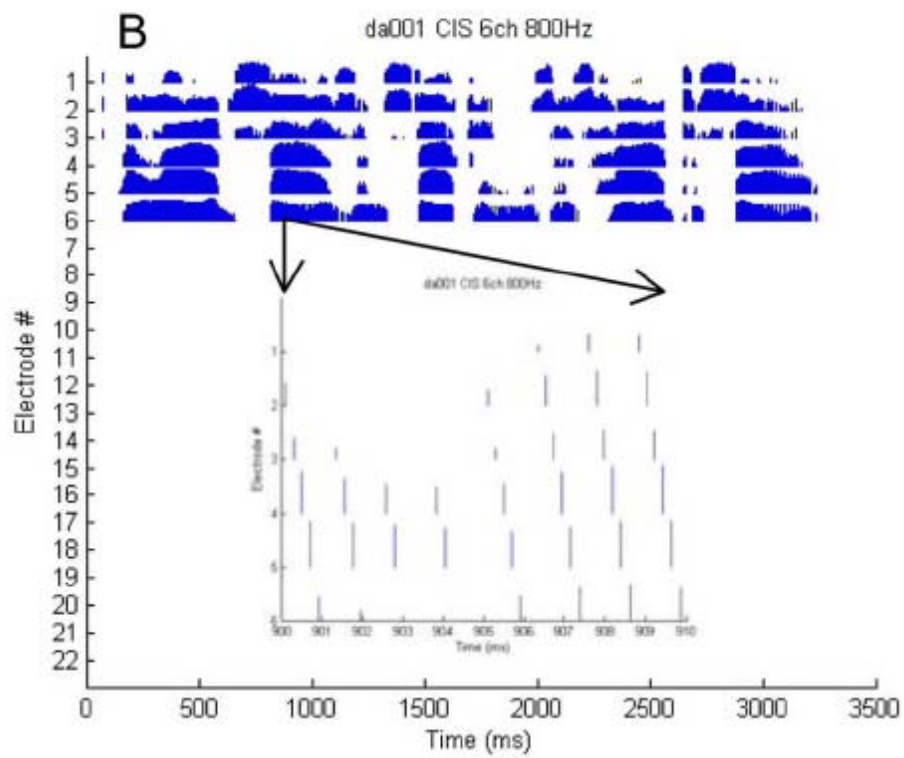
From Sounds to Electrical Pulses

Spectrogram

"A large size in stockings is hard to sell"



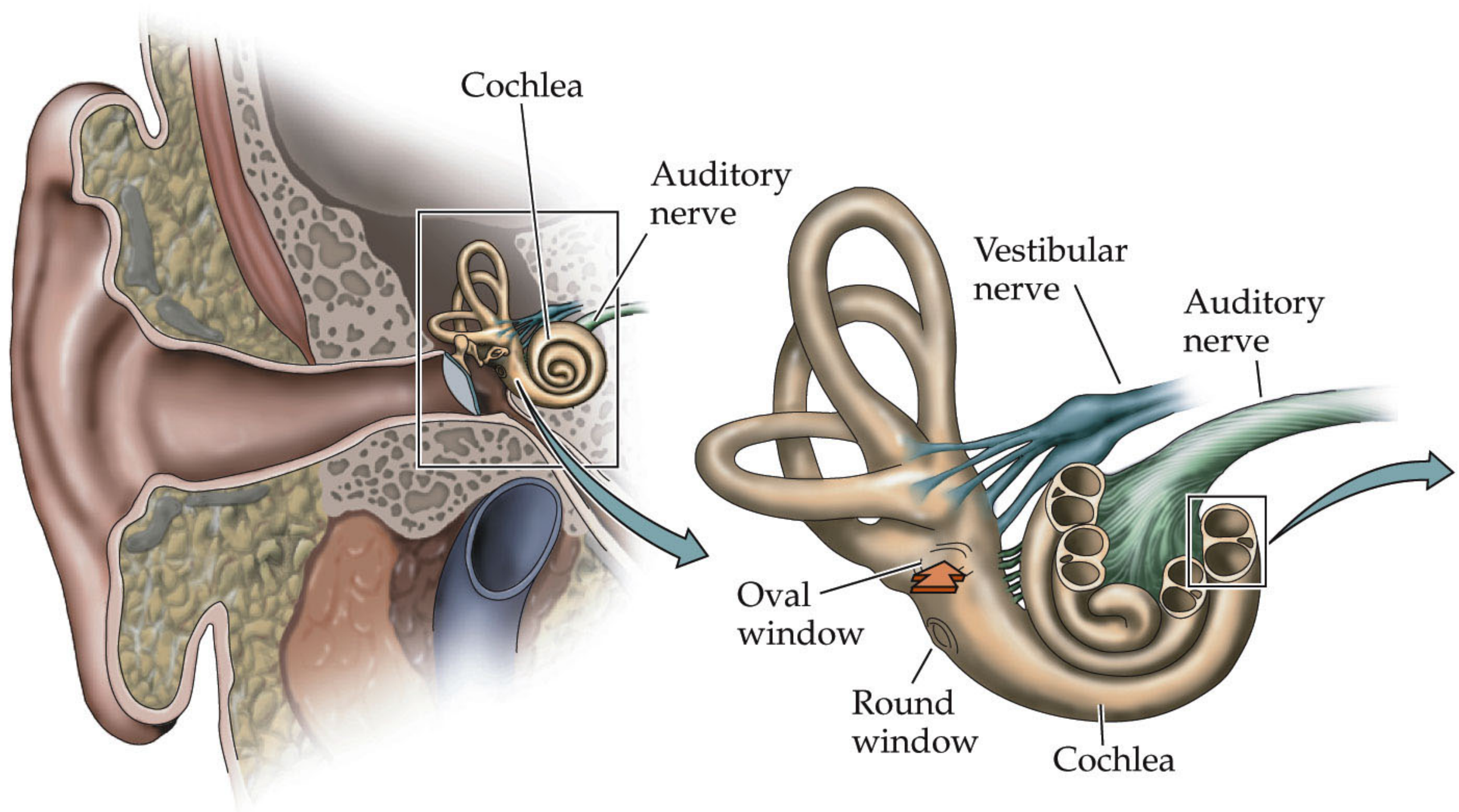
Electrodograms



Zeng (IEEE Rev BME, 2008)

How could cochlear implant work?

Cochlear implant < 24 channels. Human cochlea has >30,000 channels



NEUROSCIENCE, Fourth Edition, Figure 13.4 (Part 1)

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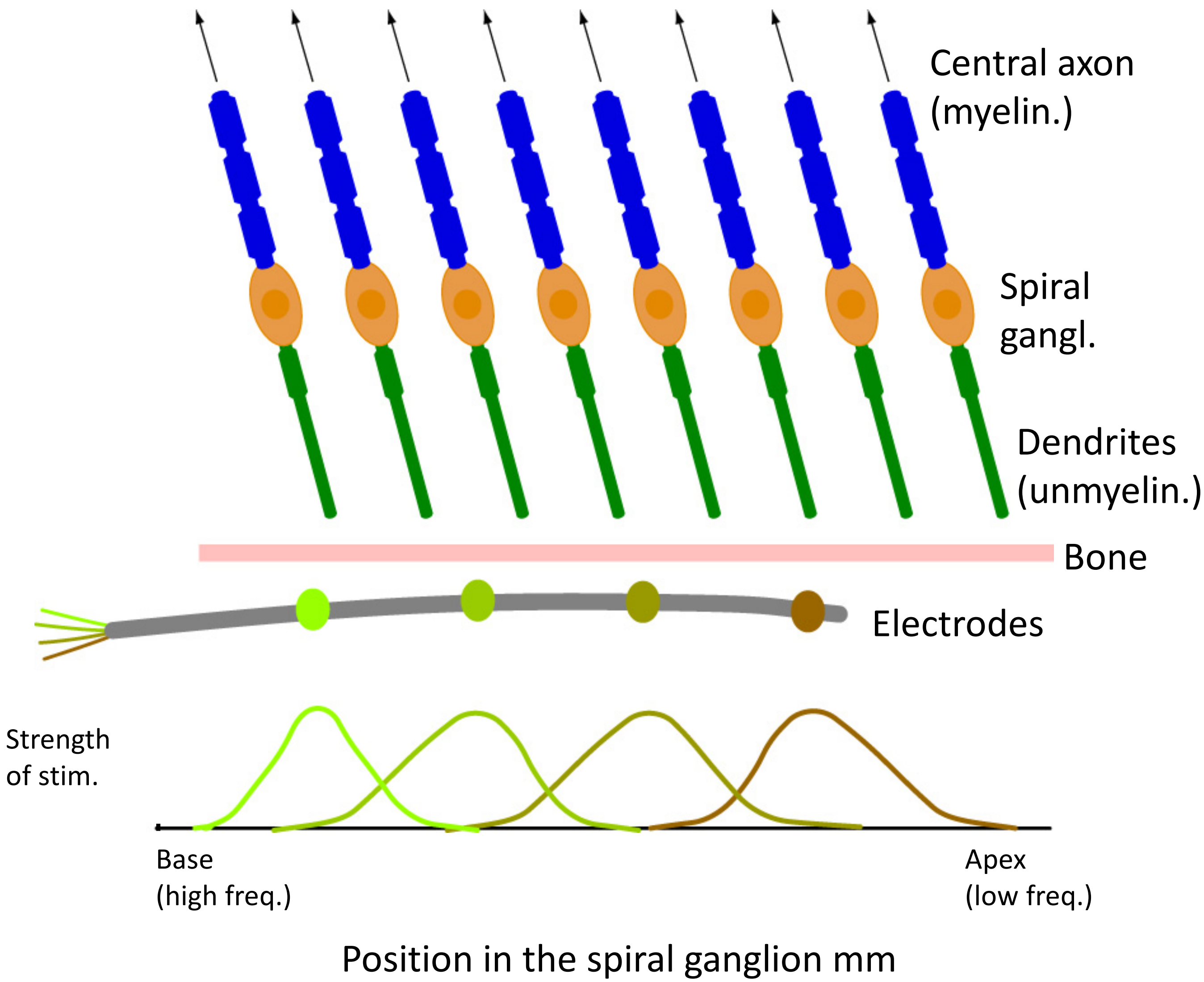
Why can't we use more channels?

The electrodes are embedded in a very conductive fluid chamber causing interactions between electrodes.



The Continuous-Interleaved Sampling (CIS) Strategy

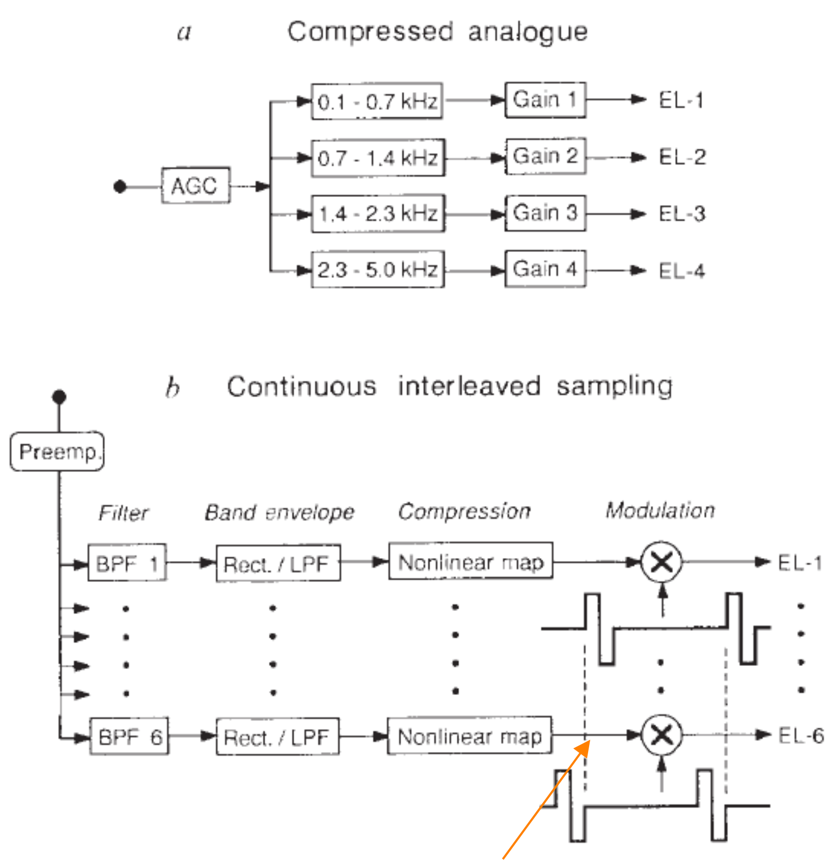
Inspired by neurophysiology of auditory nerve

LETTERS TO NATURE

Better speech recognition with cochlear implants

Blake S. Wilson*†, Charles C. Finley*†, Dewey T. Lawson*, Robert D. Wolford‡, Donald K. Eddington§ & William M. Rabinowitz§

* Neuroscience Program, Research Triangle Institute, Research Triangle Park, North Carolina 27709, USA
† Division of Otolaryngology and ‡ Center for Speech and Hearing Disorders, Duke University Medical Center, Durham, North Carolina 27710, USA
§ Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, Cochlear Implant Research Laboratory, Massachusetts Eye and Ear Infirmary, Boston, Massachusetts 02114, and Department of Otology and Laryngology, Harvard Medical School, Boston, Massachusetts 02115, USA



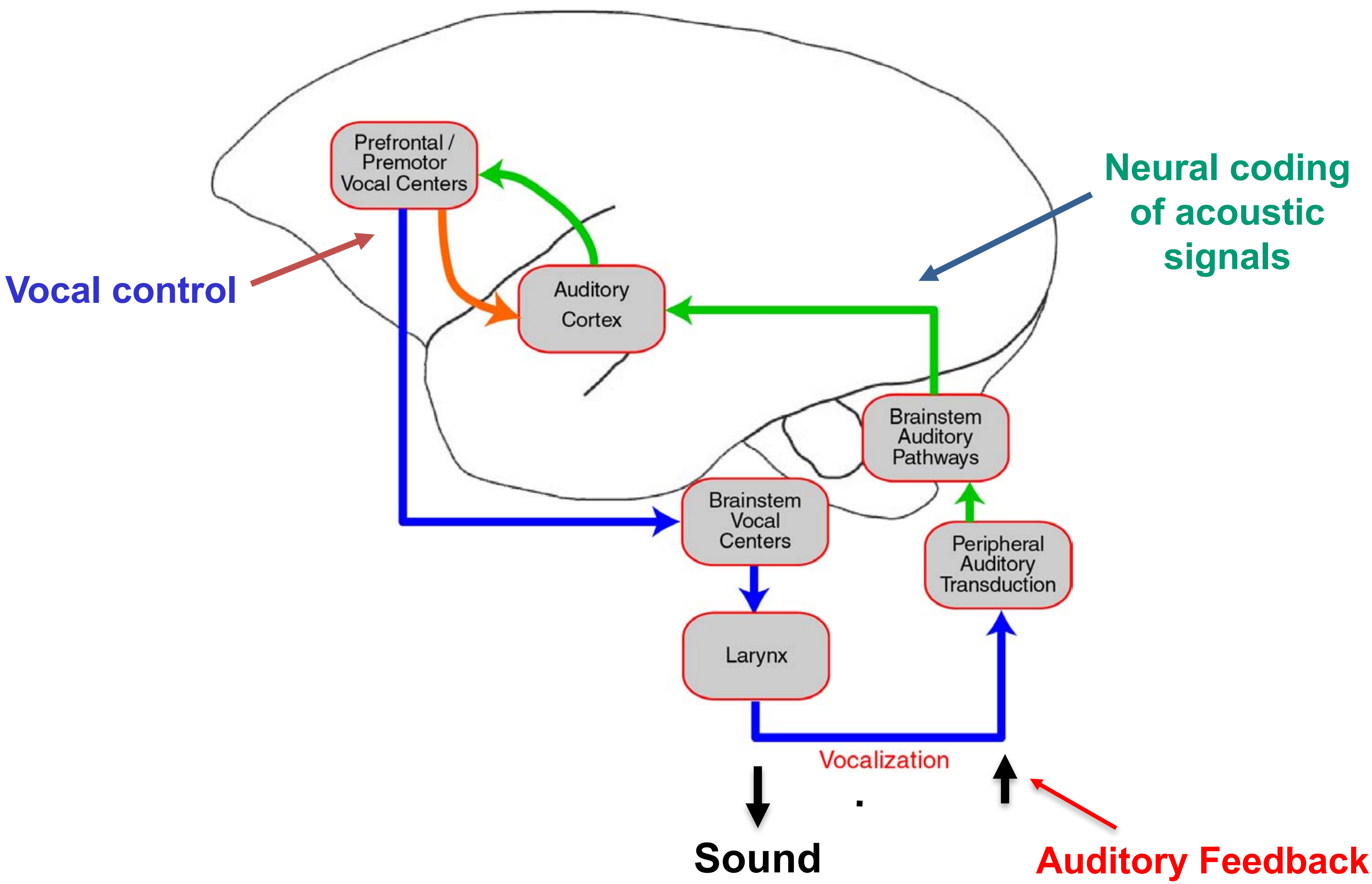
Note the non-simultaneous electrode activation!

Limitations of the Current Cochlear Implant Devices

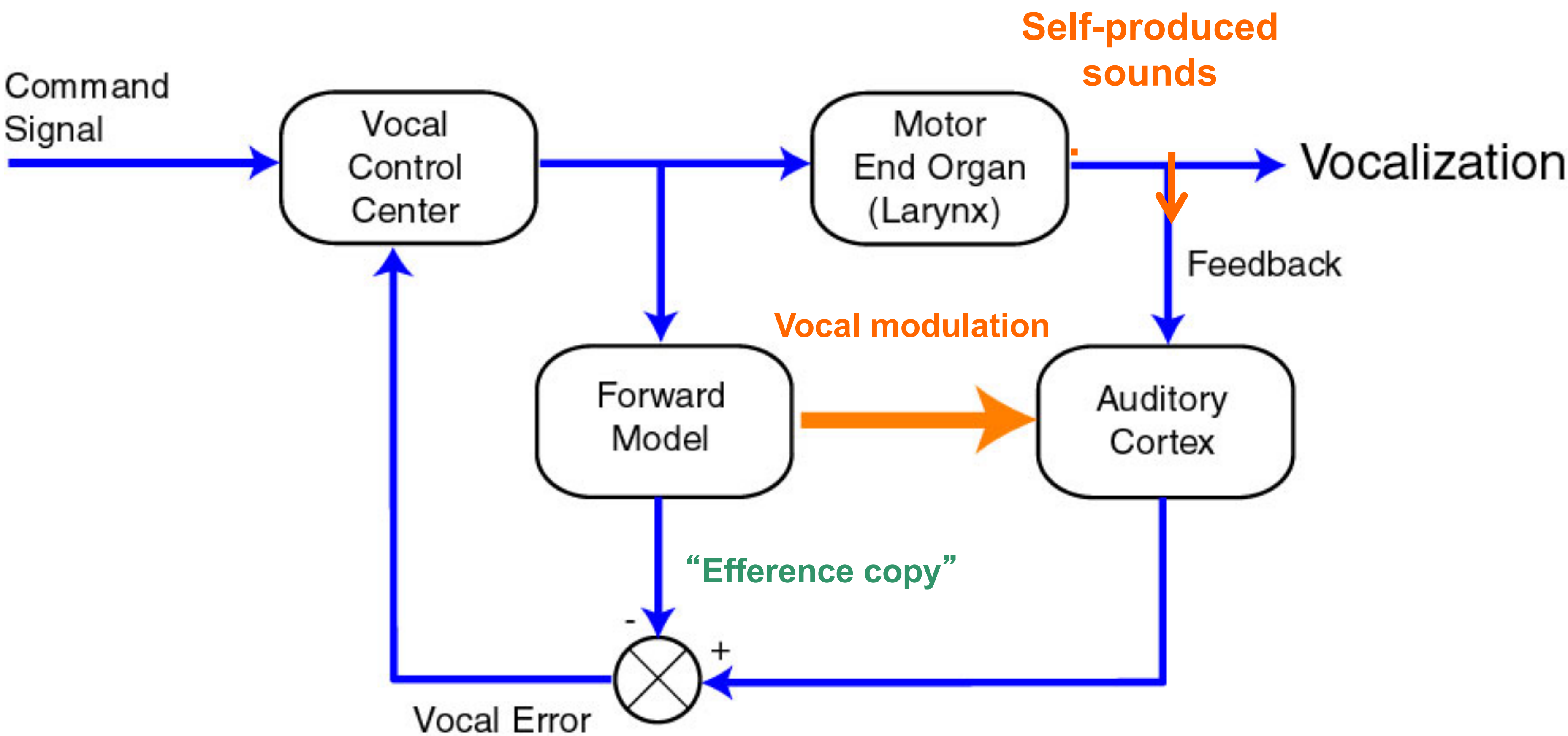
- Unexplained large variations among users
- Poor performance in noisy background
- Poor performance in tonal language and music perception
- No spatial hearing
- Lack of accurate vocal feedback

Auditory-Vocal Pathways

We listen, talk and listen while talking

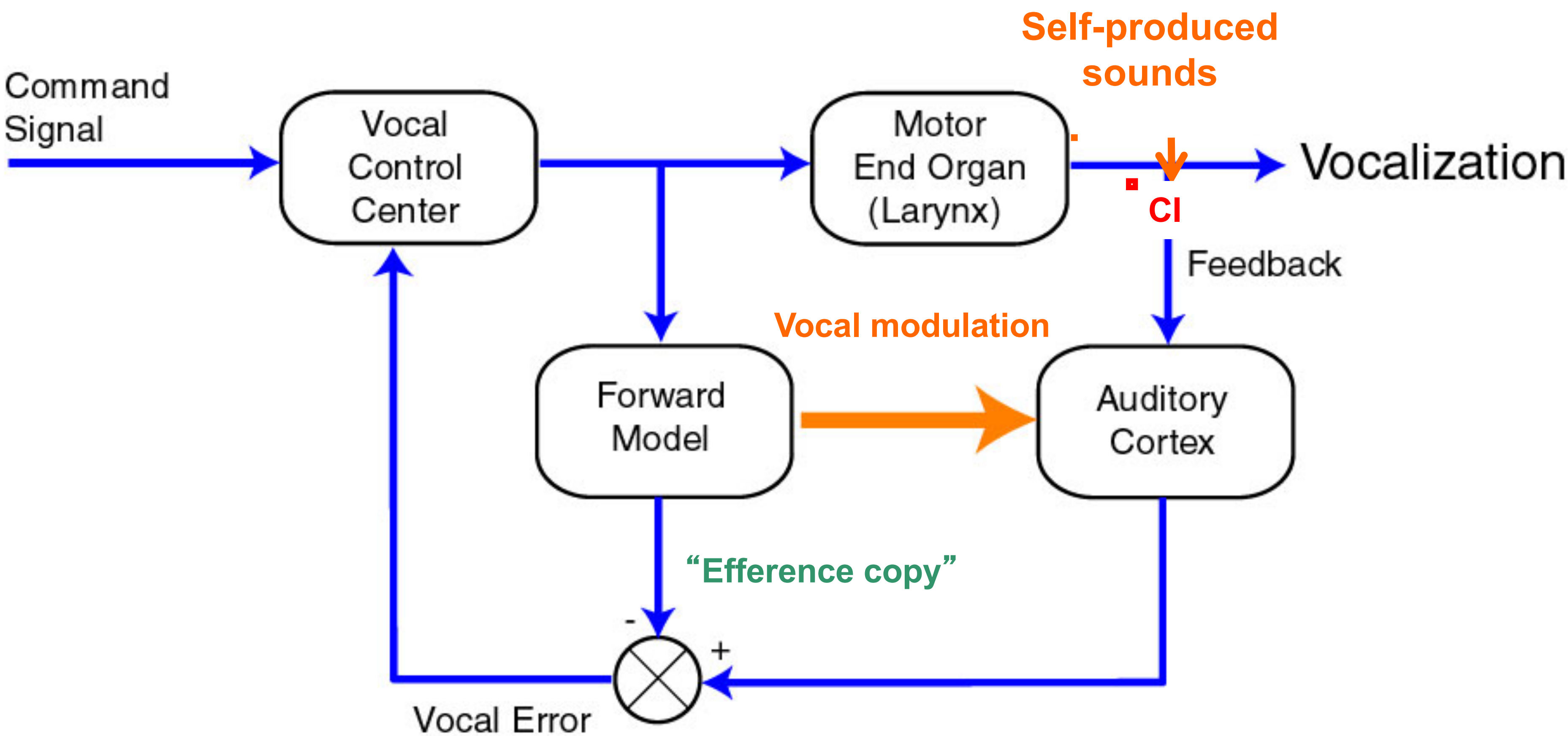


Auditory-vocal interactions in auditory cortex



Vocal error = 0 (normal feedback)

How could CI users speak accurately?

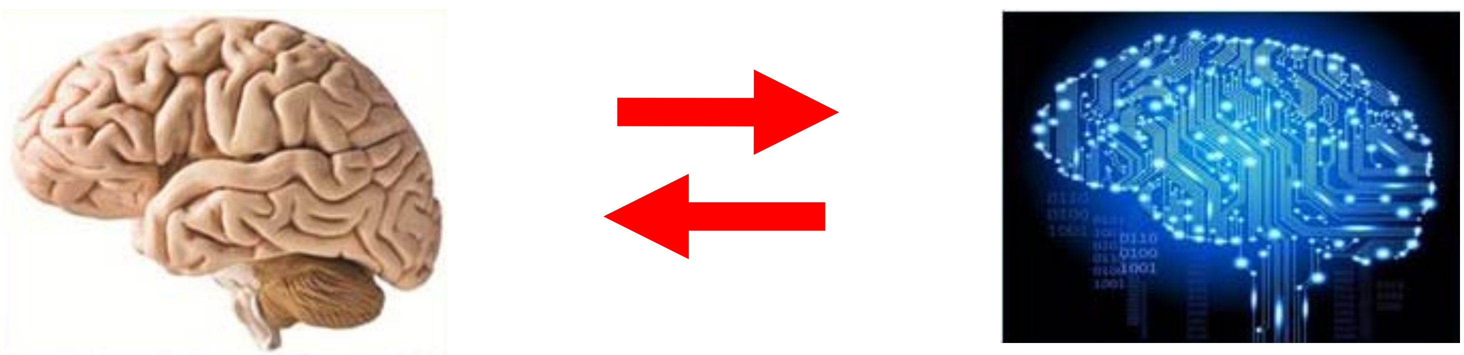


Vocal error > 0 (inaccurate feedback in CI users)

清华大学脑与智能实验室
Tsinghua Laboratory of Brain and Intelligence

智能的本质是什么？

What is the biological basis of intelligence?



理解大脑对于推动人工智能的研究具有极大的意义

清華大學 生物医学工程系

提要

- 什么是生物医学工程?
- 生物医学工程专业培养什么人才?
- 生物医学工程专业毕业生就业的前景如何?
- 生物医学工程专业的学生可以从事什么样的研究?

欢迎同学们随时提问（举手即可）

什么是生物医学工程（BME）？

Biomedical engineering is a discipline that advances knowledge in engineering, biology and medicine, and improves human health through cross-disciplinary activities that integrate the engineering sciences with the biomedical sciences and clinical practice.

运用工程技术手段探索生命系统新知识、
研发改善医疗健康的新技术

生物医学工程（BME）专业培养什么样的学生？

既精通数理及工程原理、又具有扎实的现代生命科学知识、能够在工程和生命科学跨领域工作的交叉科学人才，能够用定量的手段解决生命医学领域负杂问题的特殊人才。

BME与传统工科的不同之处是什么？

“双语教育”

能够讲两种语言，可以从事基础研究和技术开发

BME毕业生就业方向

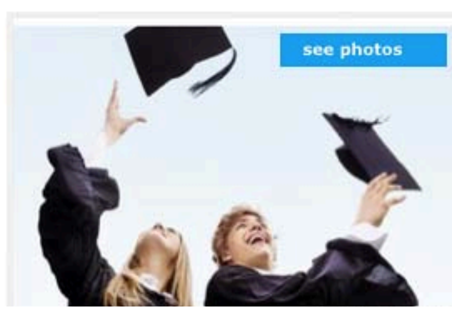
科学研究、技术开发、创业、管理

- 工科各个领域
- 生命科学研究机构
- 医疗仪器研发公司
- 政府医疗仪器审核部门

Forbes

The 15 Most Valuable College Majors

With rising tuition costs and a rapidly changing job landscape, a student's college major is more important than ever. It can either set you up for lifetime career success and high earnings or sink you into debt with few avenues to get ahead of it.



At No. 1, biomedical engineering is the major that is most worth your tuition, time and effort.

清华大学生物医学工程学科发展历史

-

- 1979 Master program in Dept. of Electrical Engineering
- 1982 Undergraduate program in Dept. of Electrical Engineering
- 1986 Ph.D. program in Dept. of Electrical Engineering
- 1998 Postdoctoral program in Dept. of Electrical Engineering
- 2002 Dept. of Biomedical Engineering established
- 2017 入选"双一流学科"

瞄准国际前沿的基础研究与
响应国家需要的应用研究齐头并进

研究方向

神经工程
Neural Engineering

医学影像
Medical Imaging

微纳医学 / 组织工程
Micro-Nano Medicine
Tissue Engineering

医疗仪器
Medical Devices

电子/生医大类培养

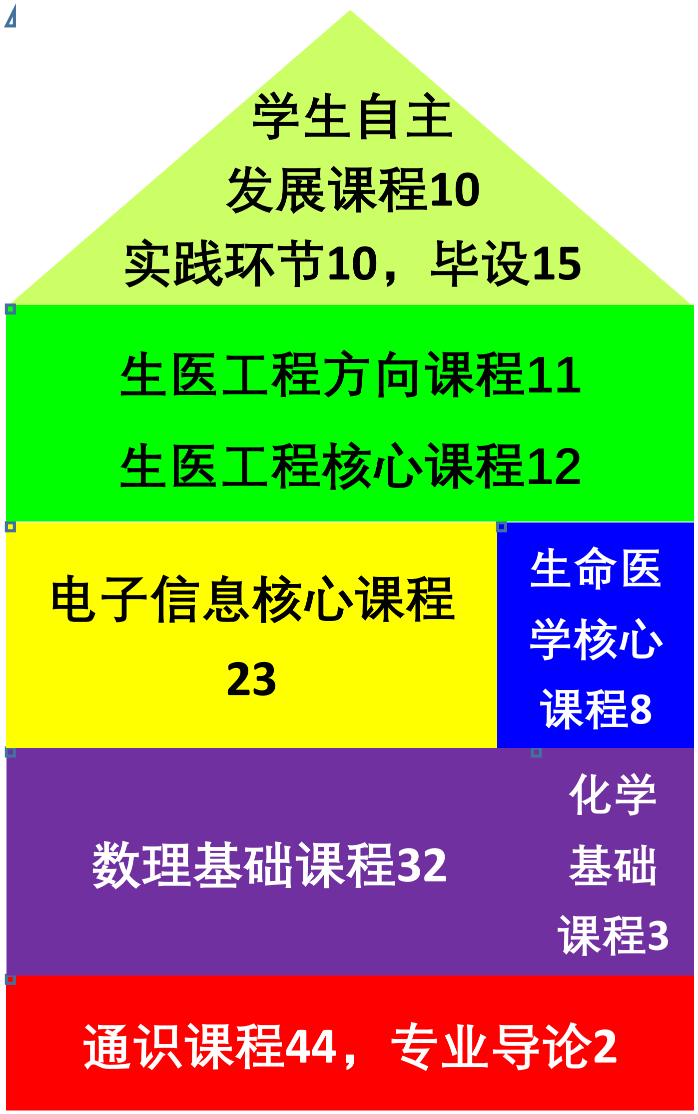
大类招生与大类培养不能分离，本科阶段要突出**通识教育，注重全面培养**，特别是学生综合素质能力的提升，培养终身教育和终身学习的良好习惯。学校要**以学生为中心，以宽口径的大类培养为方向**，以学生成长为学校教育教学目标，**坚持大类招生与大类培养的初心不改**。

电子/生医大类培养（个人解读）

1. **完全开放电子、生医**两系现有的所有课程
2. 同学**自主选择**两系培养计划
3. 允许培养过程中同学**自主调整**培养计划
4. 增设**辅修**，给学有余力的同学更宽广的学习机会
5. **“双下标”**，让同学在电子系和医学院享受更丰富的培养机遇和资源，形成更广泛的情感纽带

（正式官方解读请参加下周一晚的宣讲会）

生医系（电子方向）大类培养方案简介



电子信息类（170学分）

其中**通识**（44学分）、**数理基础**（29学分）以及**电子信息核心课程**（23学分）和电子系培养方案完全一致

电子系和生医系大类培养方案简介

第一学年完全一致

第二学年可自由替代

电子系培养方案	生医系培养方案	互相替代方案
复变函数与数理方程（3学分）	有机化学B （3学分）	
数据与算法（3学分）	概率论与数理统计（3学分）/随机数学方法（3学分）二选一	可替代（生医）
电子信息科学与技术导引（2）（1学分）		
概率论与随机过程（1）（2学分）	生物化学原理（4学分）	
电磁场与波 （3学分）/电动力学 （4学分）二选一	人体解剖与生理学（4学分）	可作为生医自主发展课程10
物理电子学基础实验 （1学分）		可作为电子自主发展课程》21

生医工程辅修学位（申请中）

生物医学工程辅修专业修业年限2年，总学分不少于25学分

主要课程：**必修课：4门，16学分**

课程名称	学分	时间	备注
人体解剖与生理	4	大二下	必修
信息与生命	4	大三上	必修
生物医学检测原理与传感技术	4	大三上	必修
生物医学电子学	4	大三下	必修

生医工程辅修学位（申请中）

- 生物医学工程辅修专业修业年限2年， 总学分不少于25学分

主要课程：**限选课：不少于9学分，在下述课程中选修不少于3门**

大三学年及大四上学期开设

医学影像（1）	3学分
医学影像（2）	3学分
医学图像	4学分
神经科学与工程原理	3学分
系统与计算神经科学	3学分
神经建模与数据分析	3学分
生物医学材料基础	3学分
组织工程学原理	3学分
生物芯片技术及其应用	3学分

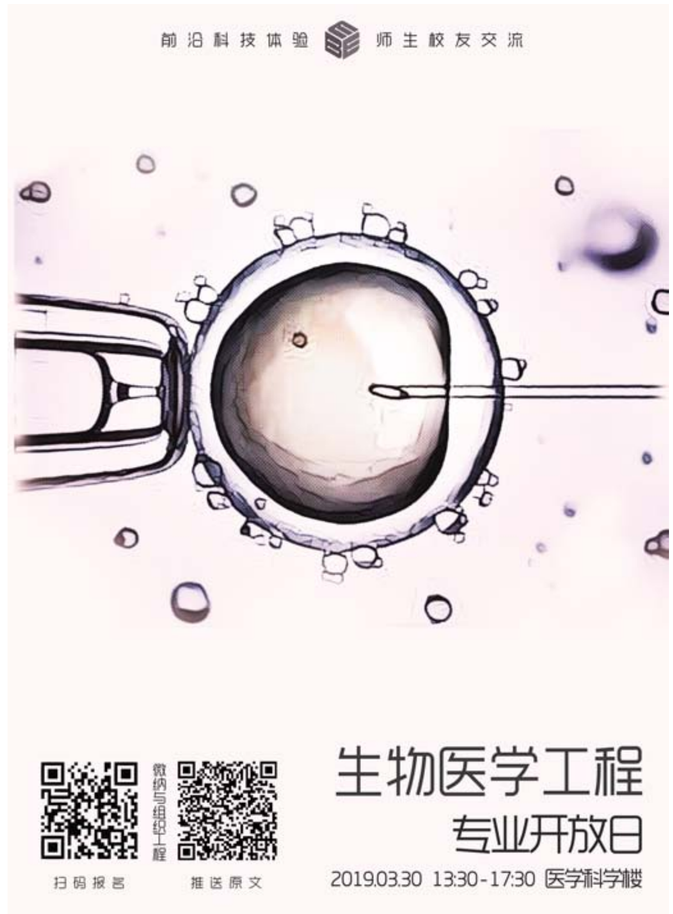
欢迎参加生物医学工程专业开放日

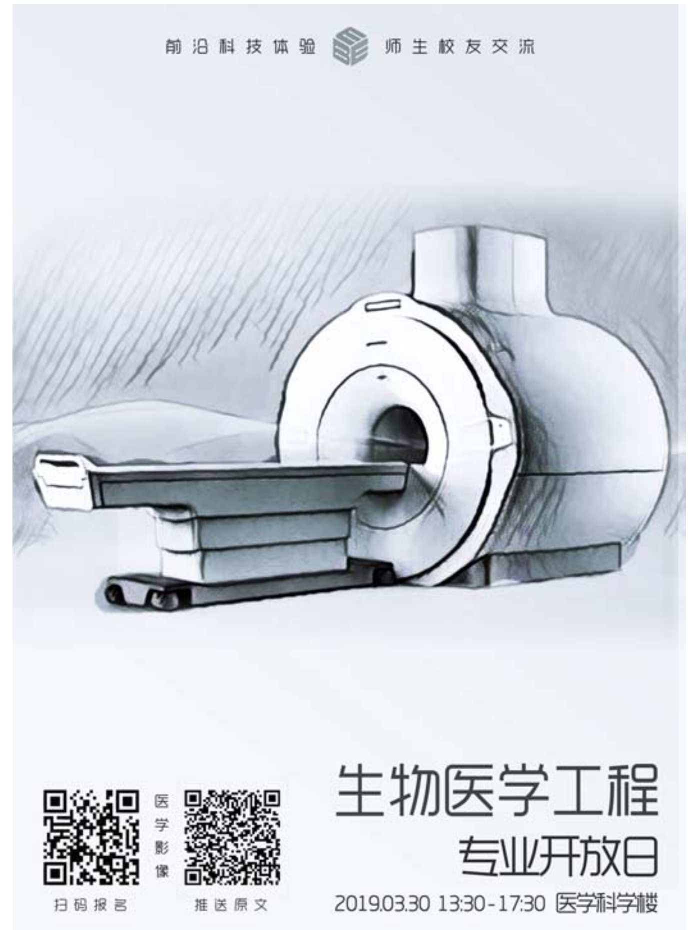
3月30日（周六）下午13:30-17:30

医学科学楼二层

欢迎参加生物医学工程专业开放日

3月30日（周六）下午13:30-17:30　医学科学楼二层





- 丰富精彩体验环节
- 精美小纪念品
- 美味茶歇

欢迎参加生物医学工程专业开放日

生物医学工程专业宣讲咨询会

30日周六17:30 医学科学楼C201

开放日之后

欢迎同学们积极参加！